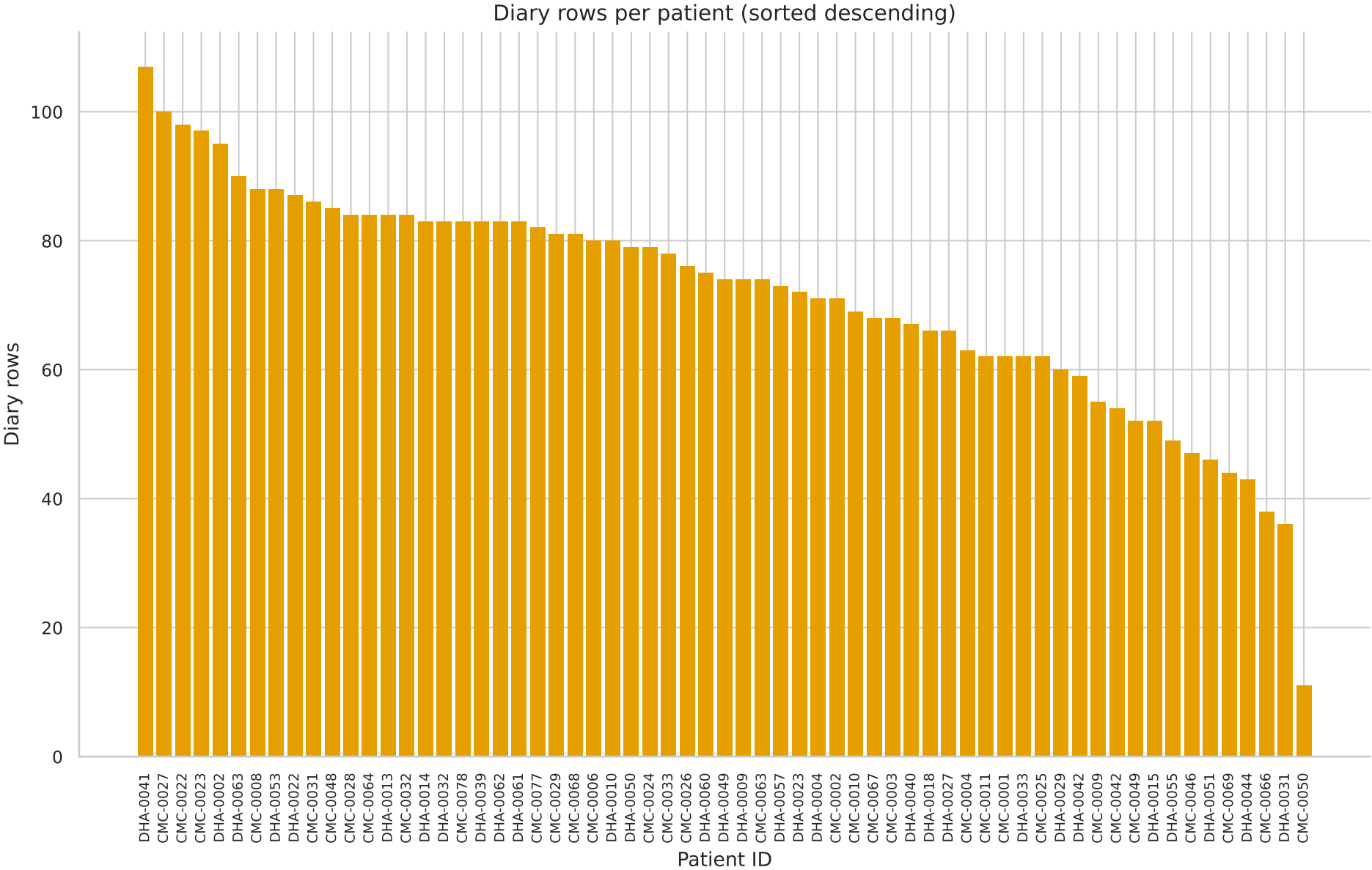


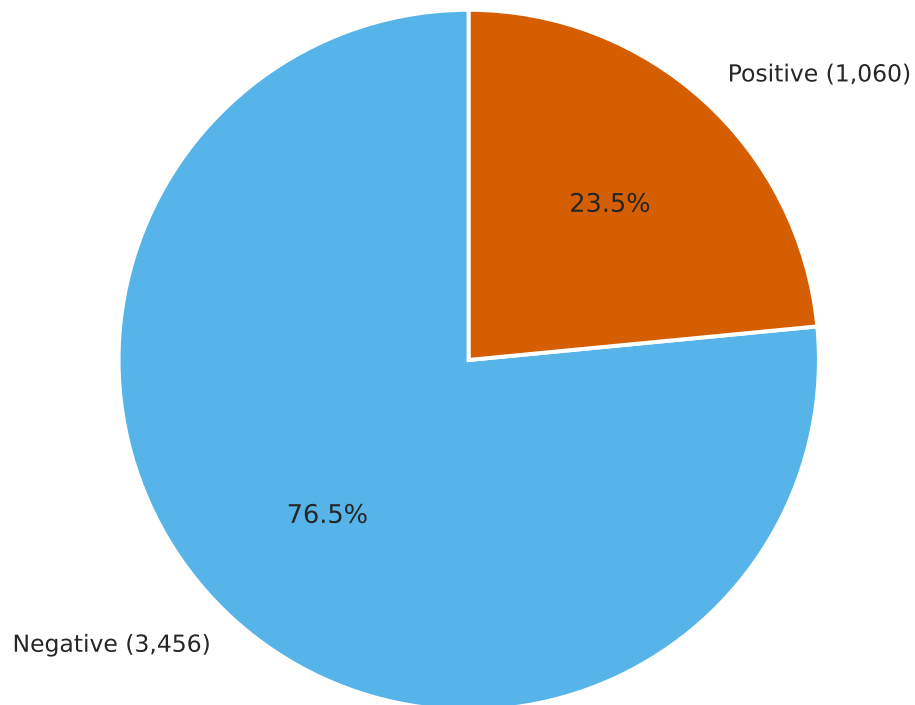
Dataset Analysis — headache

Metric	Value
Total rows	4,516
Total patients	63
Date range	2014-08-13 → 2015-04-13
Positive rate	23.5% (1,060 events)
Feature columns	56

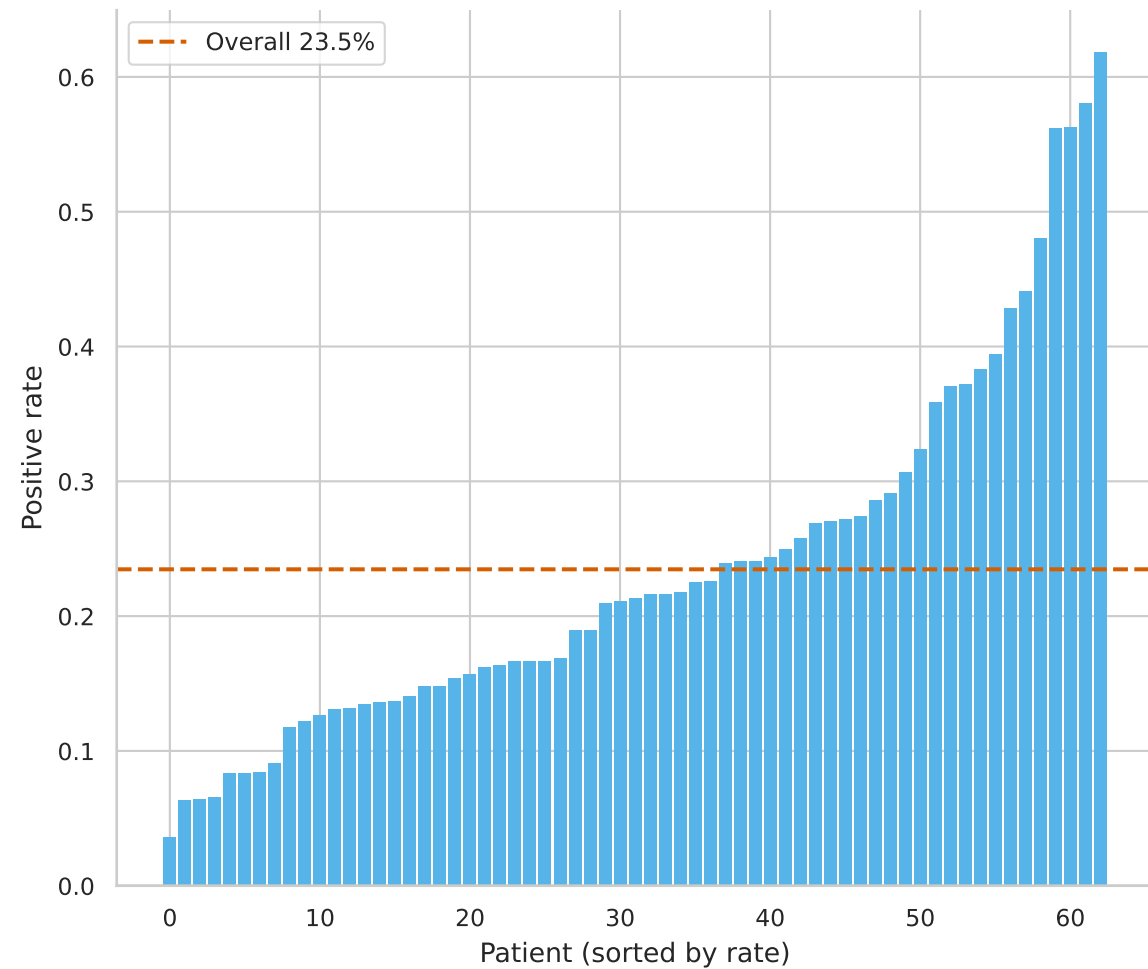


Target Distribution — headache (overall rate: 23.5%)

Overall class split

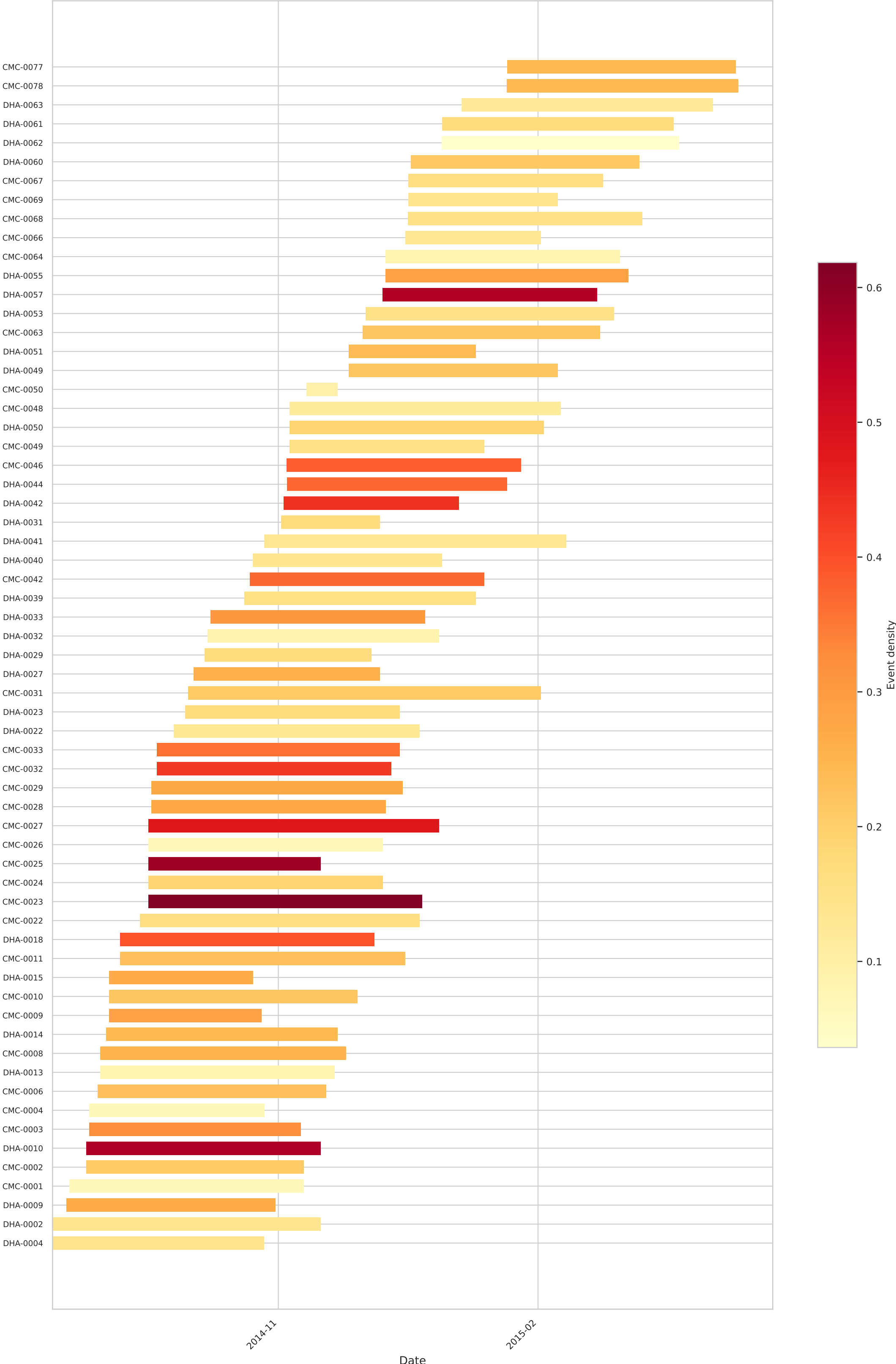


Per-patient positive rate
min=3.6% median=21.3% max=61.9%



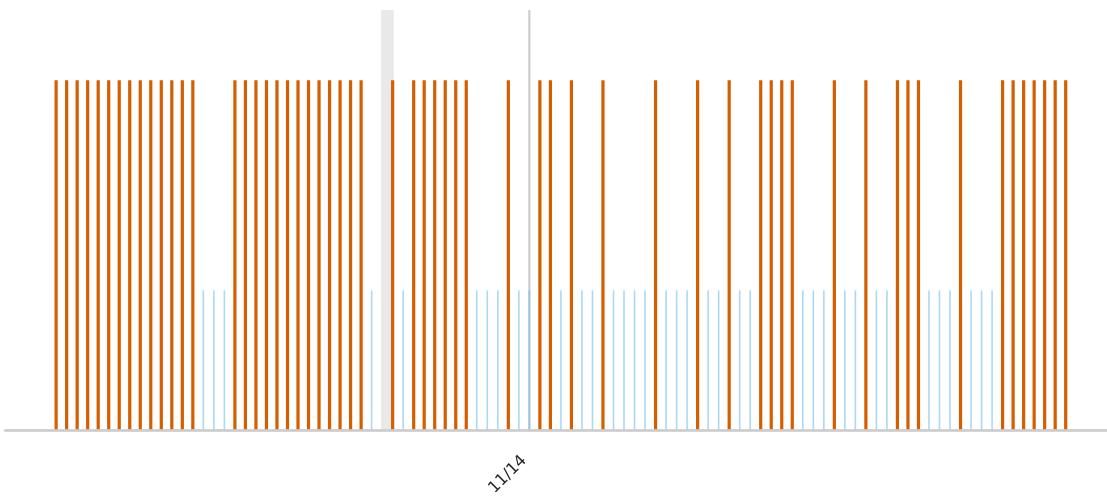
Cohort Timeline — headache

Horizontal bar = diary coverage; colour = event density

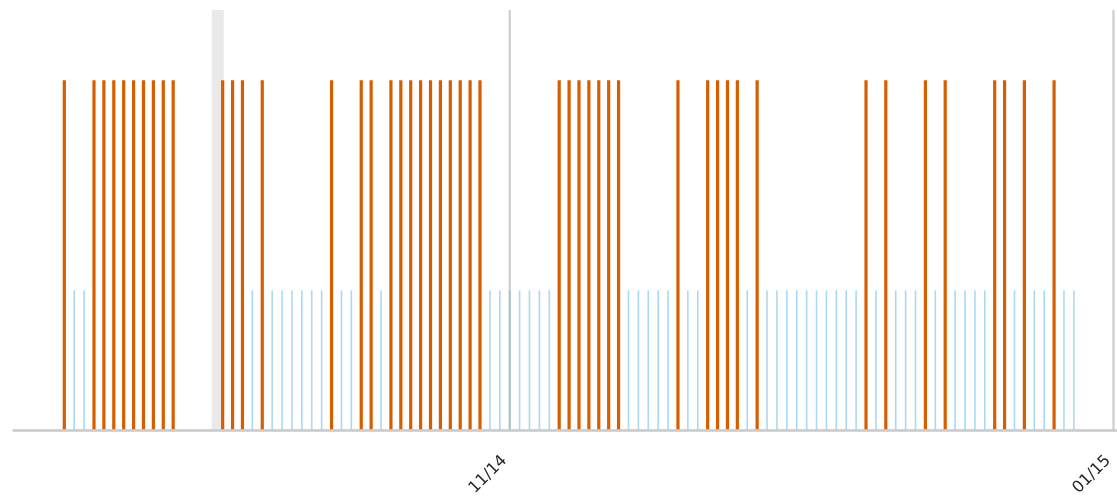


Per-patient Event Timelines — headache (patients 1-6)

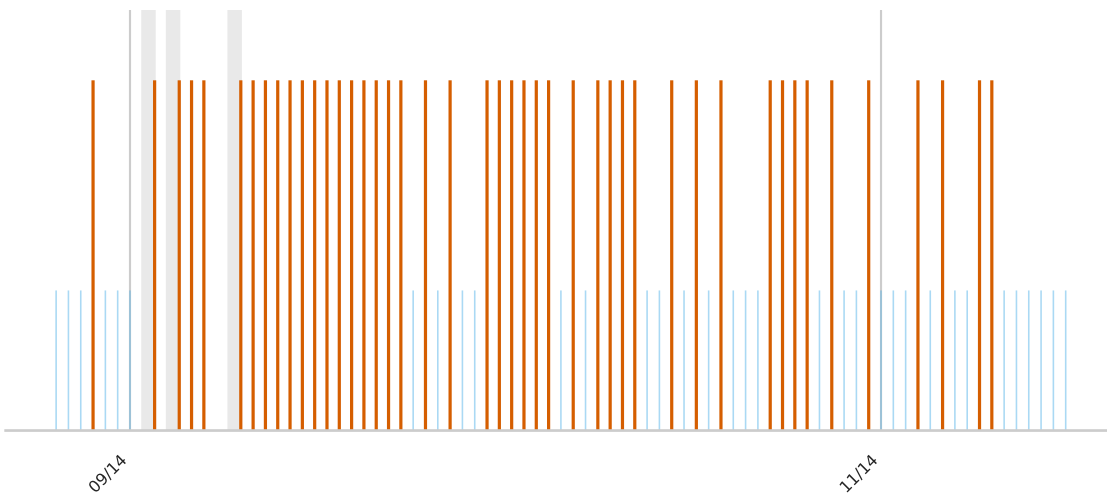
CMC-0023 | 59 events | 61.5%



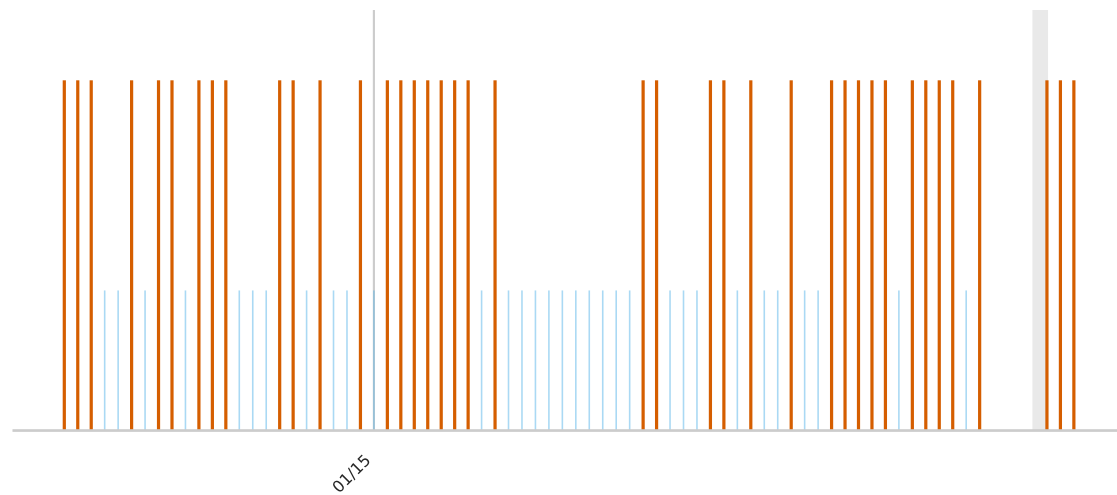
CMC-0027 | 48 events | 48.5%



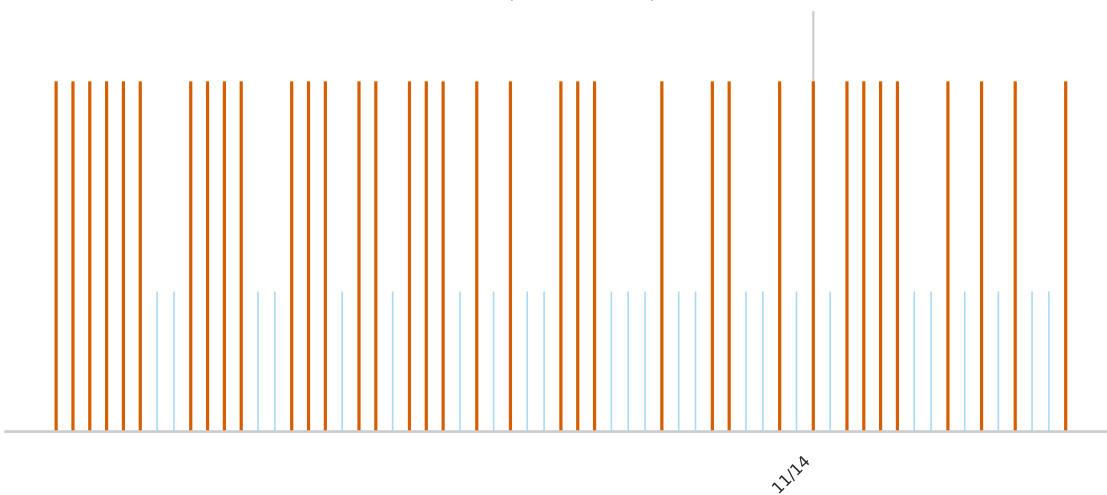
DHA-0010 | 45 events | 57.0%



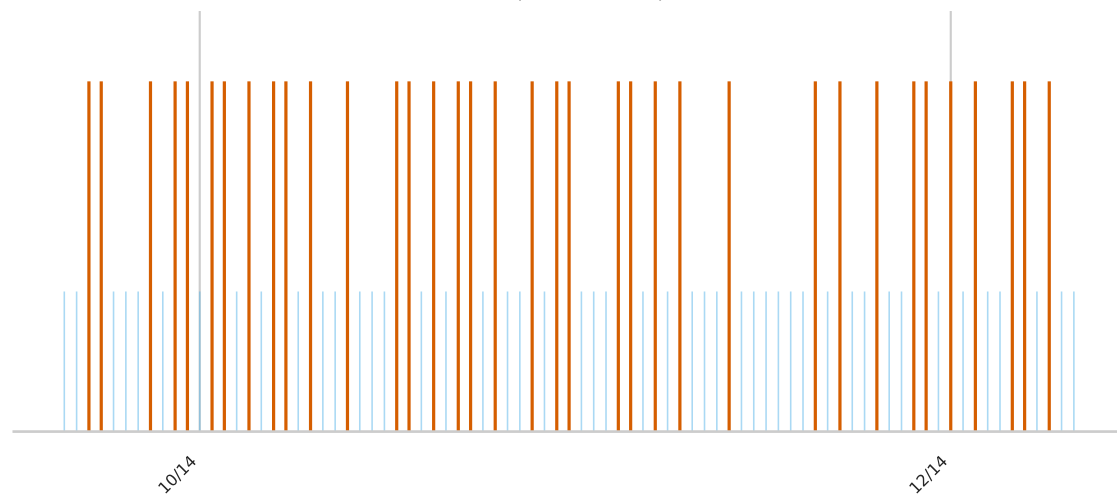
DHA-0057 | 40 events | 55.6%



CMC-0025 | 36 events | 59.0%

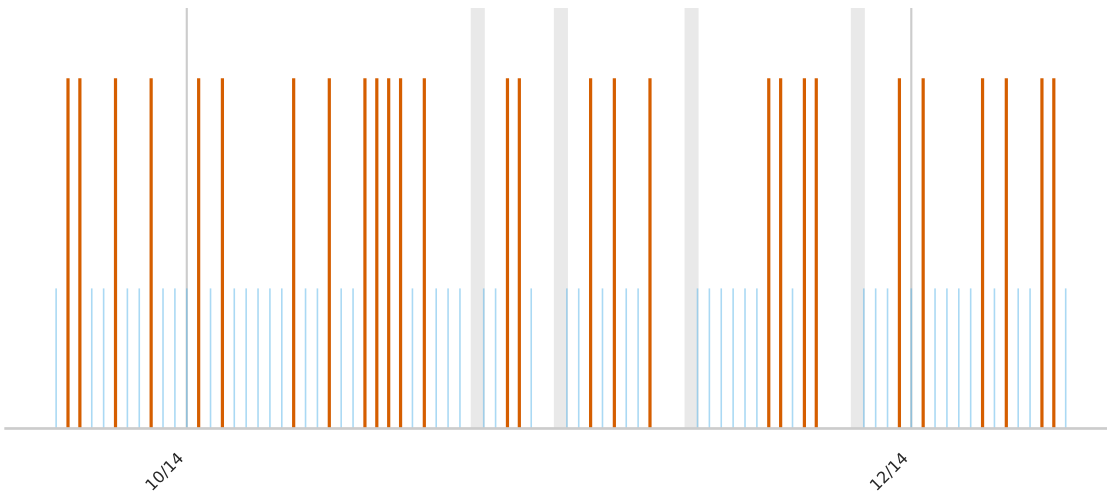


CMC-0032 | 36 events | 43.4%

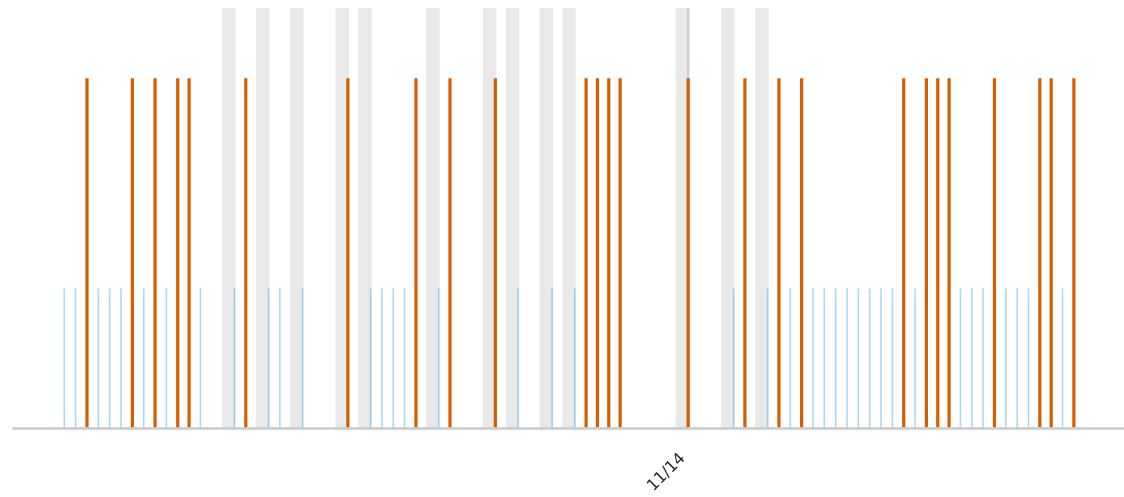


Per-patient Event Timelines — headache (patients 7-12)

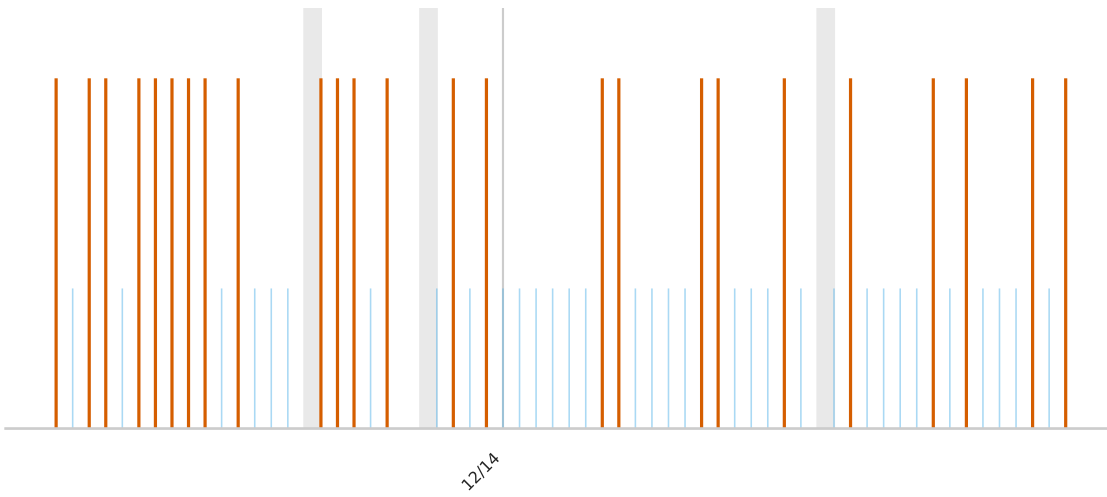
CMC-0033 | 28 events | 36.4%



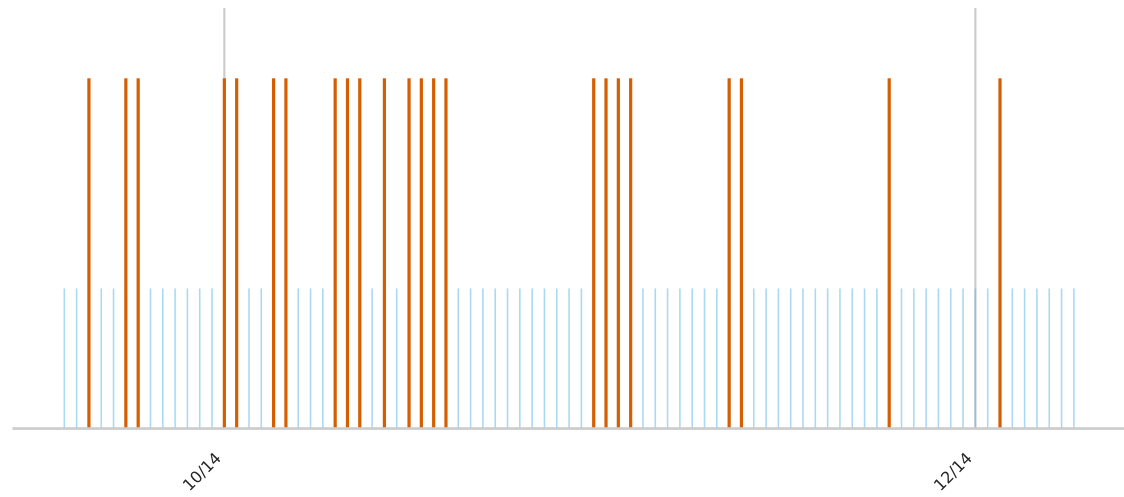
DHA-0018 | 26 events | 40.0%



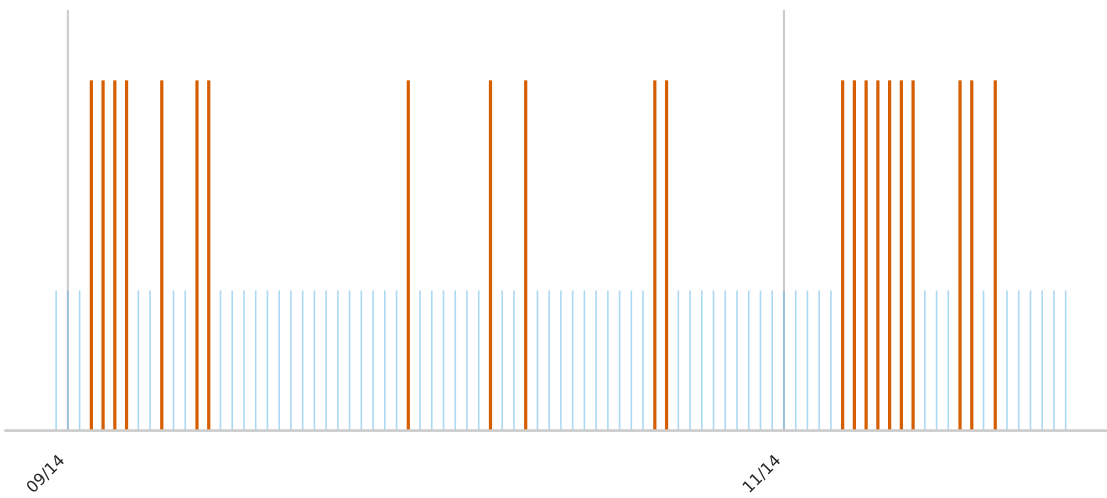
DHA-0042 | 25 events | 43.1%



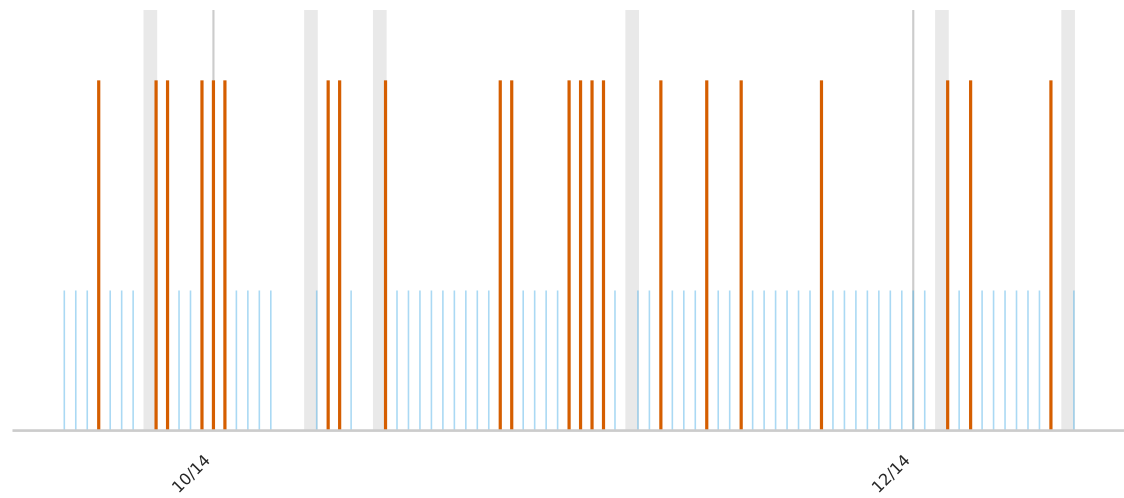
CMC-0028 | 23 events | 27.7%



CMC-0008 | 22 events | 25.3%

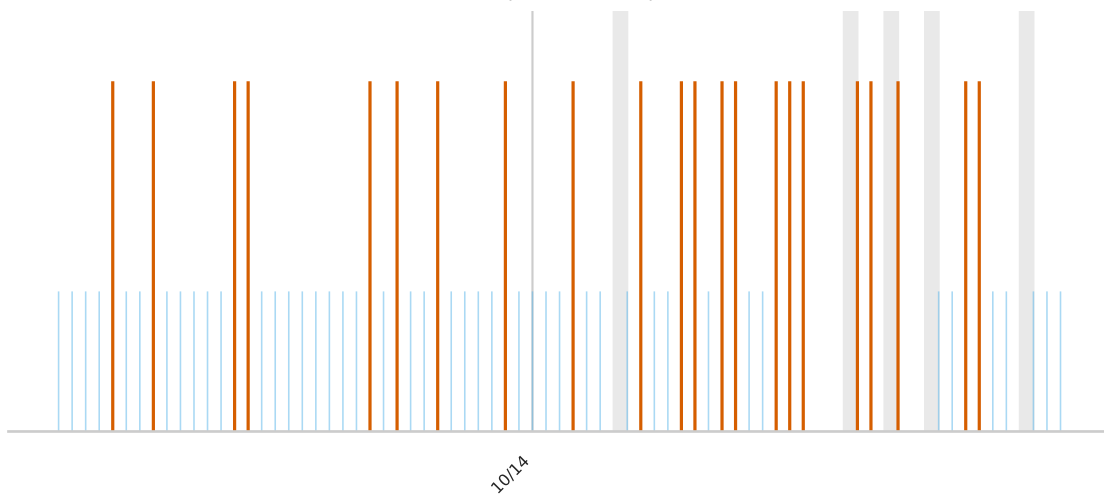


CMC-0029 | 22 events | 27.5%

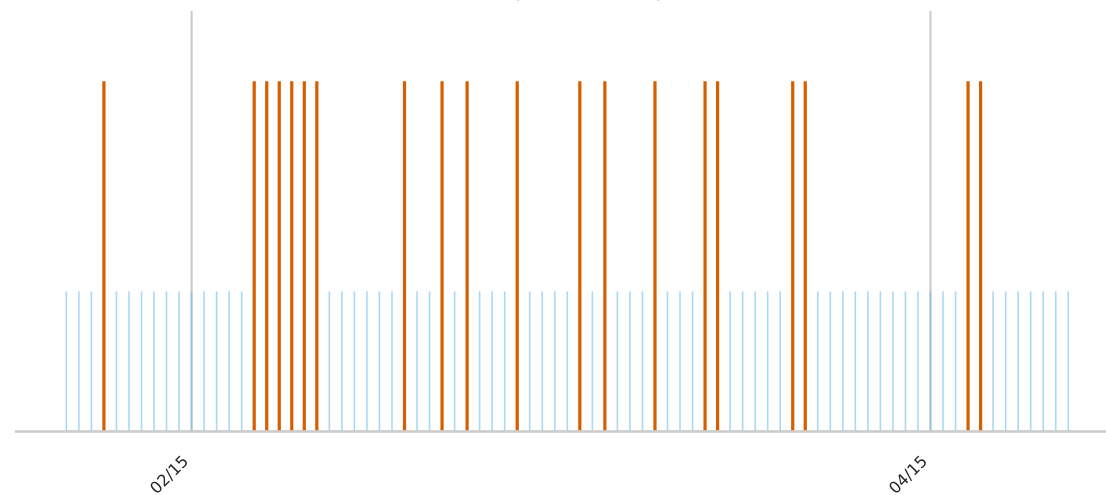


Per-patient Event Timelines — headache (patients 13-18)

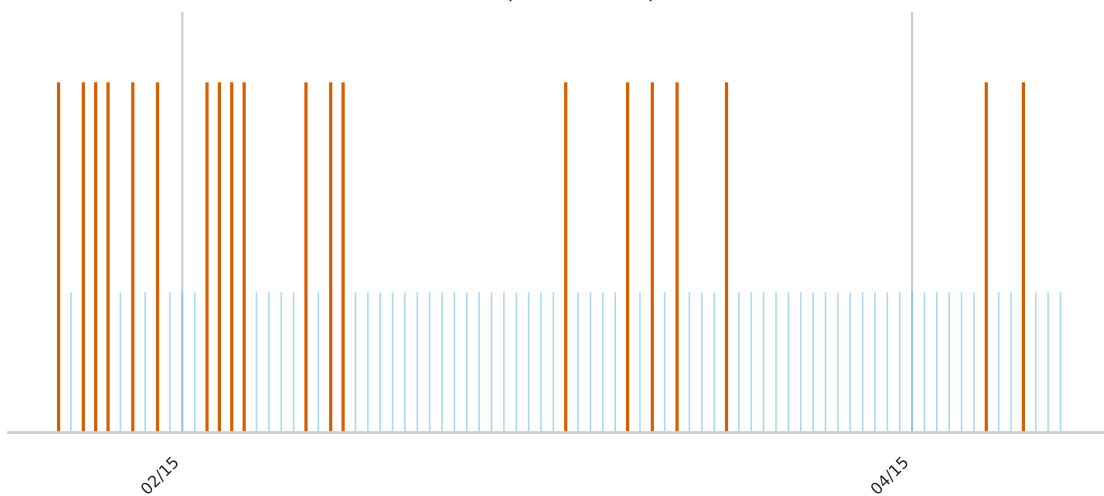
CMC-0003 | 22 events | 32.8%



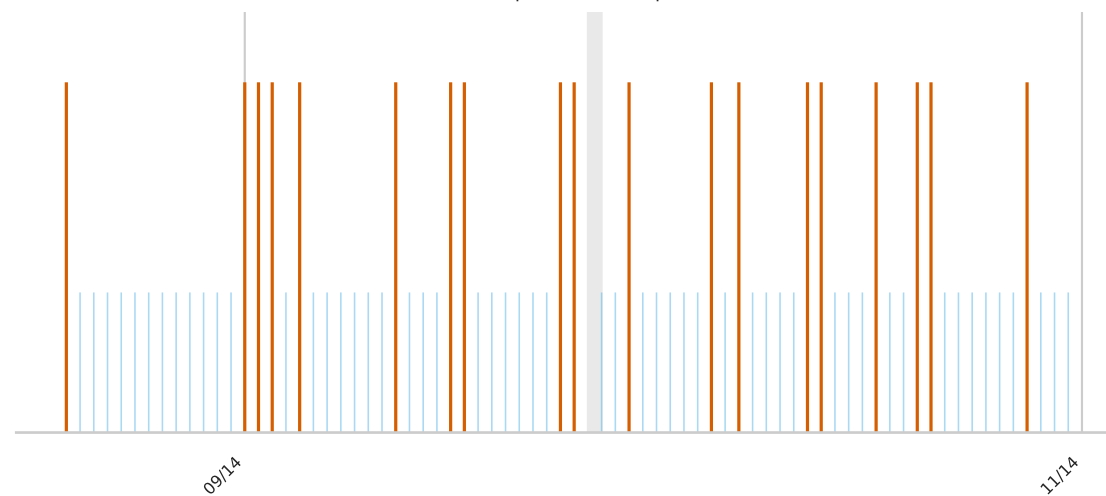
CMC-0077 | 20 events | 24.7%



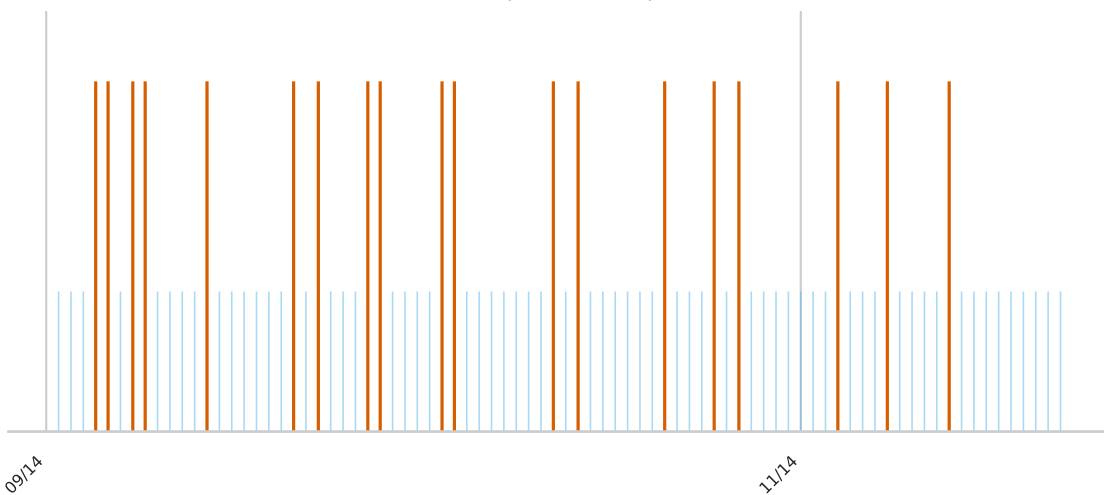
CMC-0078 | 20 events | 24.4%



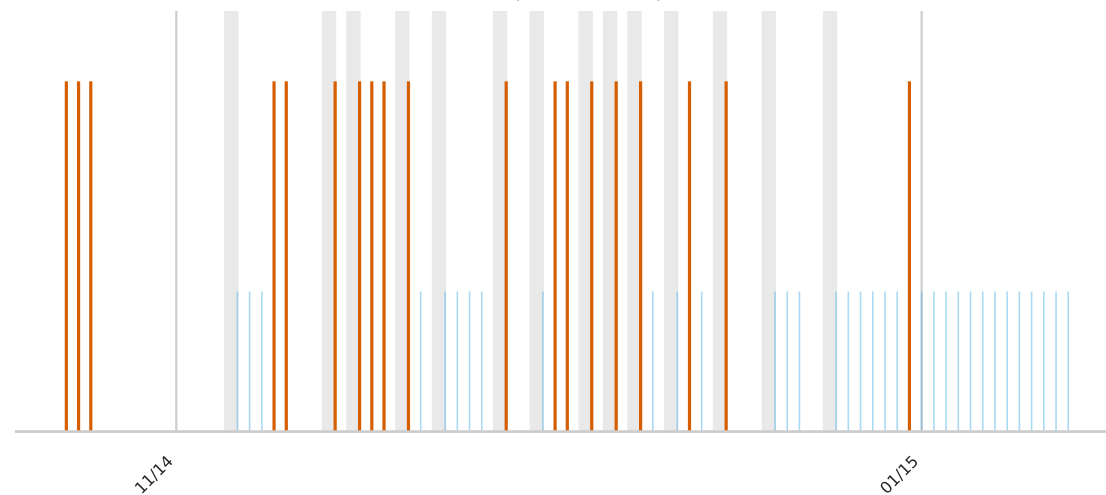
DHA-0009 | 19 events | 26.0%



DHA-0014 | 19 events | 23.2%

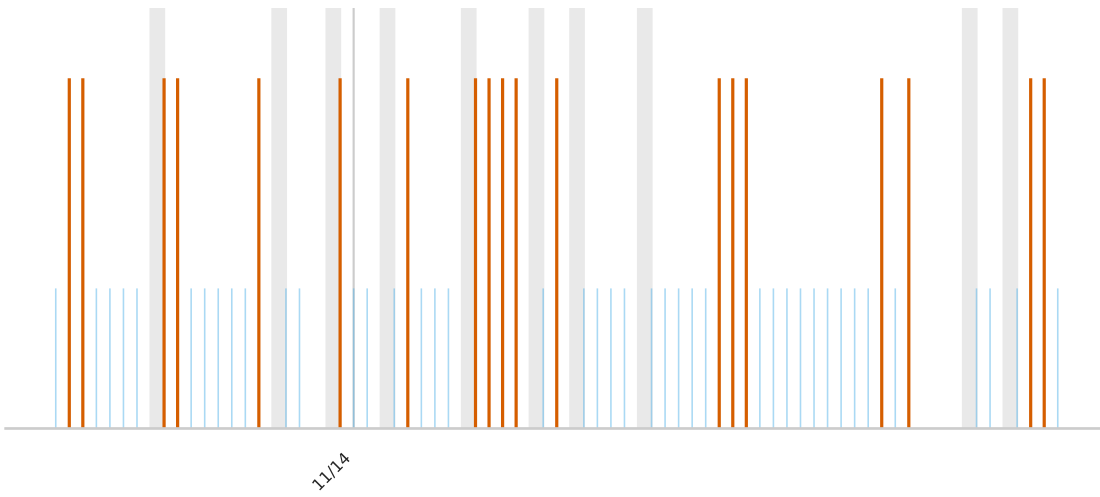


CMC-0042 | 19 events | 35.8%

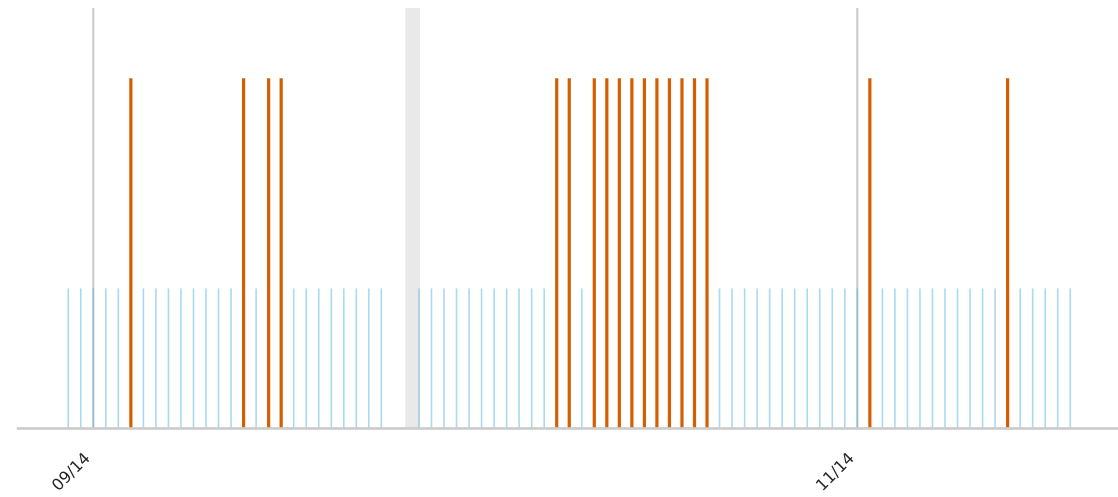


Per-patient Event Timelines — headache (patients 19-24)

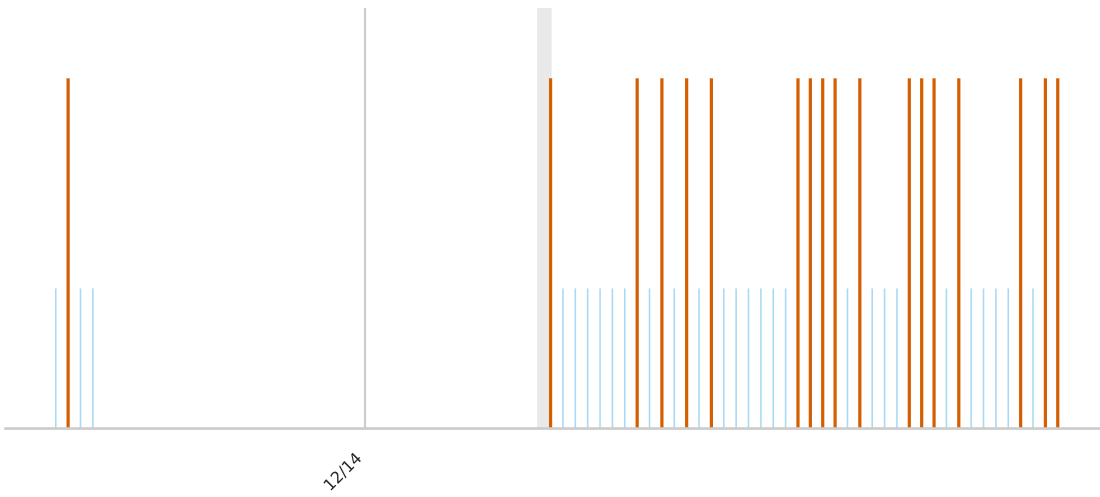
DHA-0033 | 19 events | 31.1%



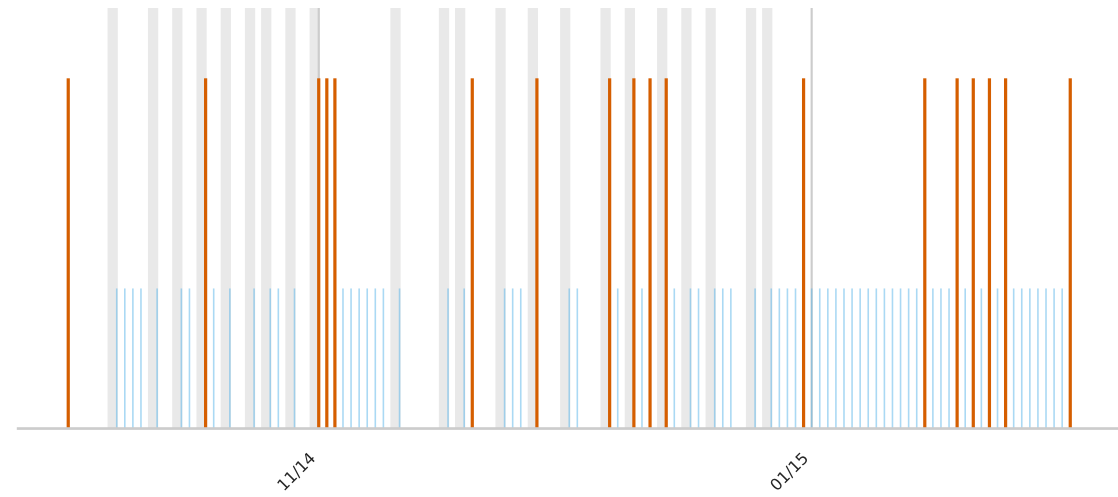
CMC-0006 | 18 events | 22.8%



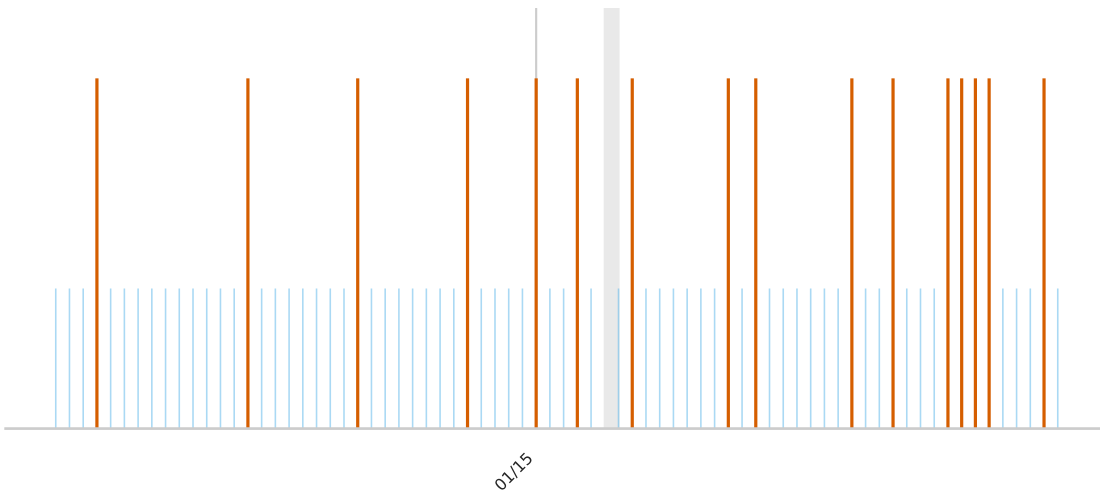
CMC-0046 | 18 events | 39.1%



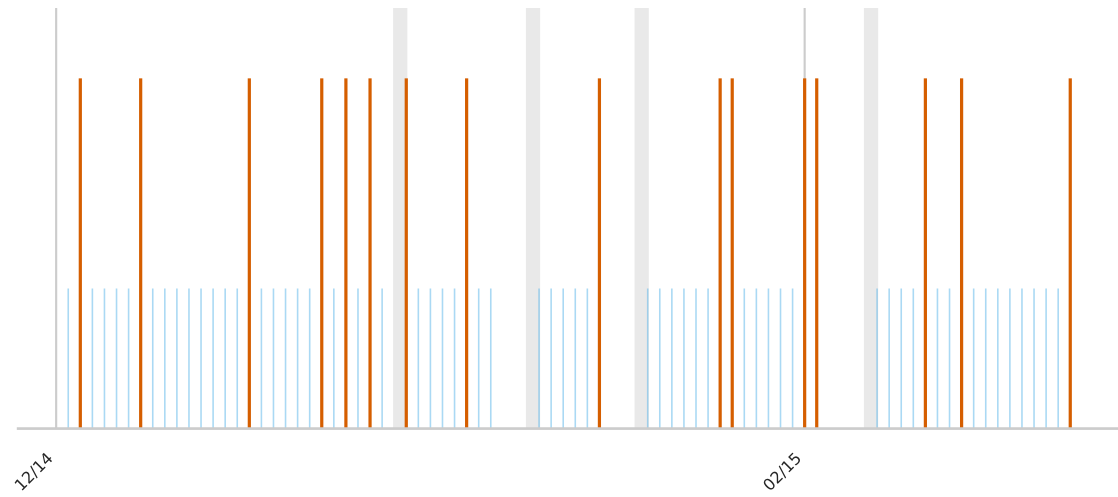
CMC-0031 | 18 events | 21.2%



DHA-0049 | 16 events | 21.9%

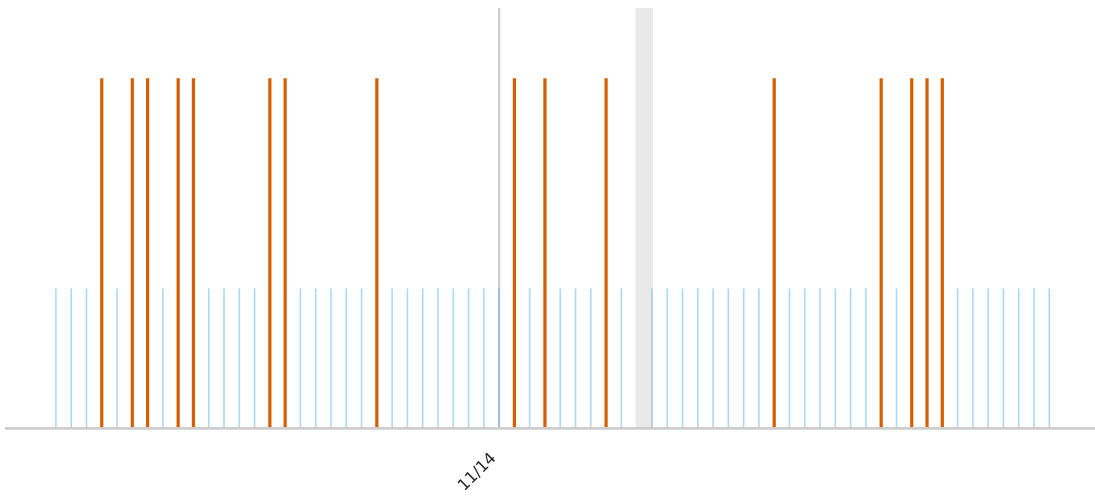


CMC-0063 | 16 events | 21.9%

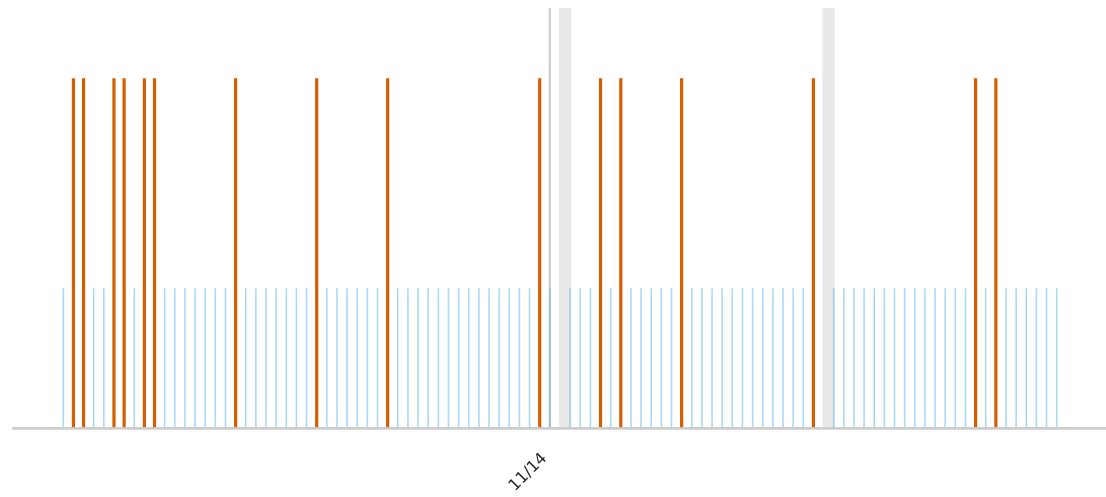


Per-patient Event Timelines — headache (patients 25-30)

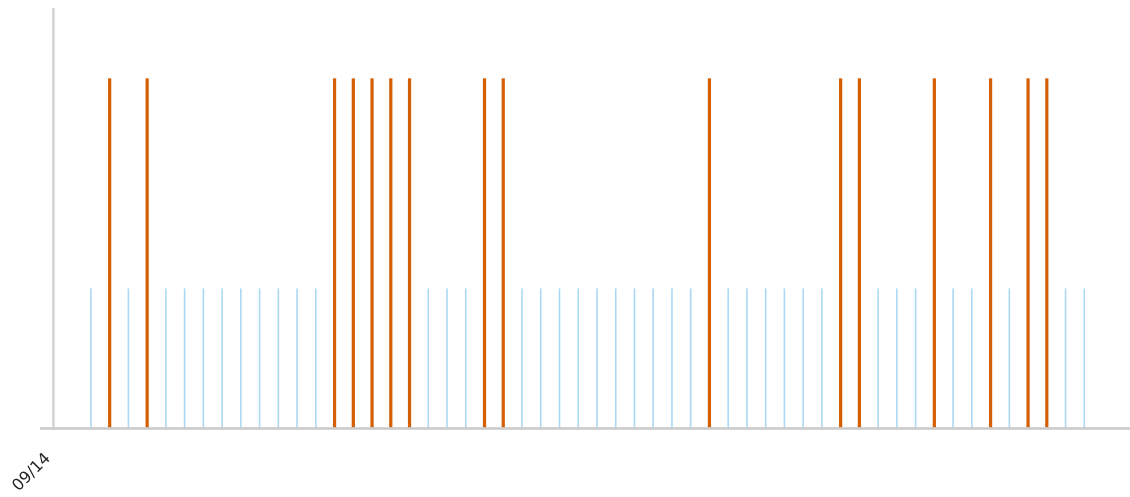
DHA-0027 | 16 events | 24.6%



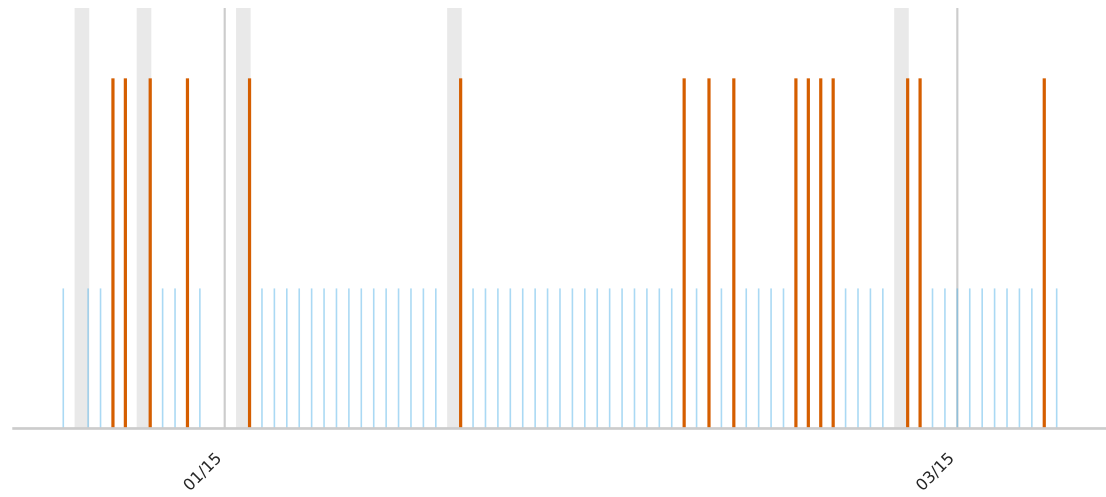
CMC-0022 | 16 events | 16.5%



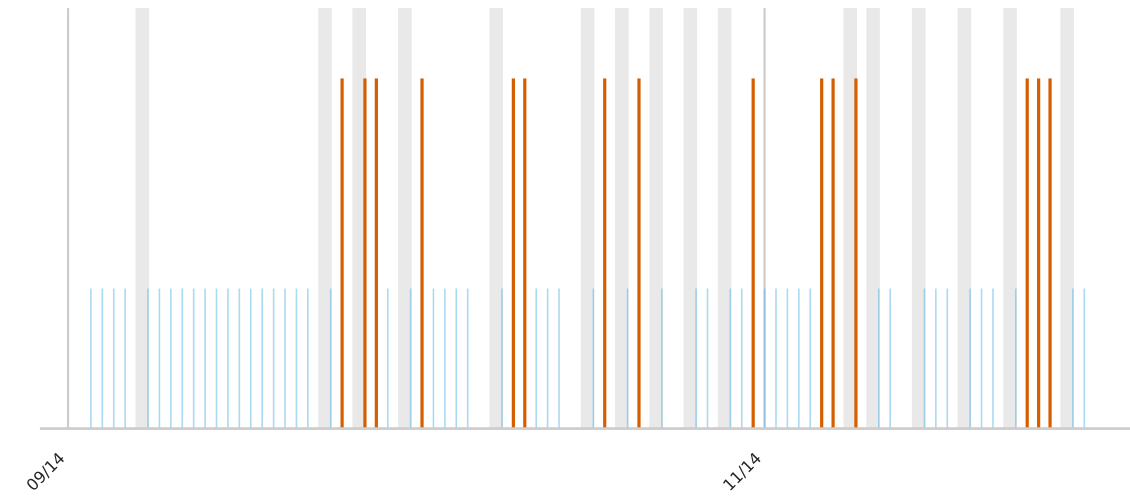
CMC-0009 | 16 events | 29.6%



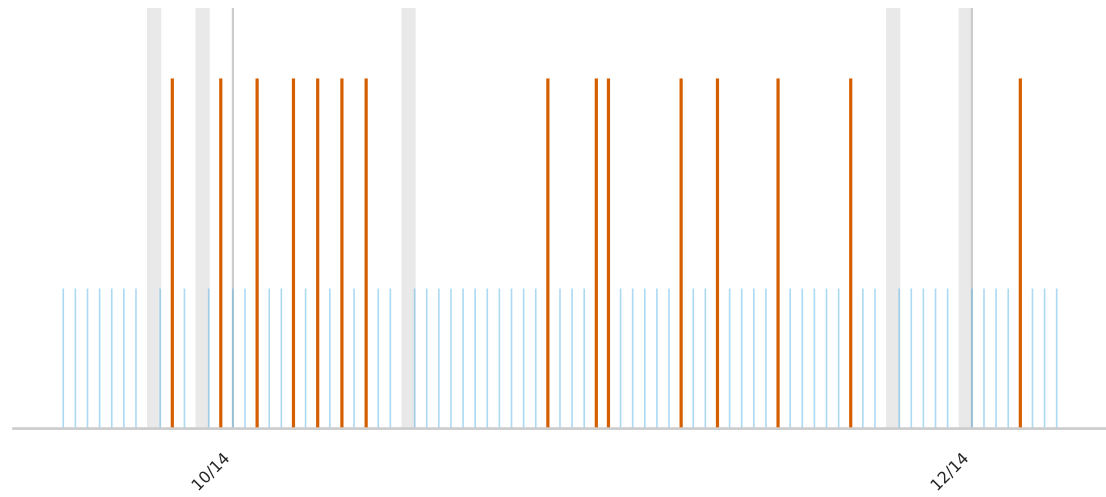
DHA-0060 | 16 events | 21.6%



CMC-0010 | 15 events | 22.1%

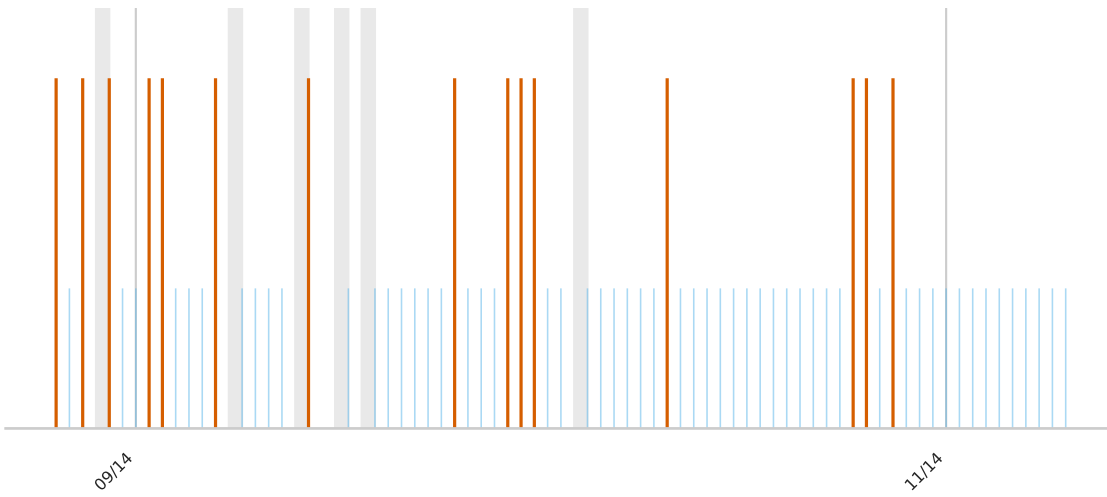


CMC-0024 | 15 events | 19.2%

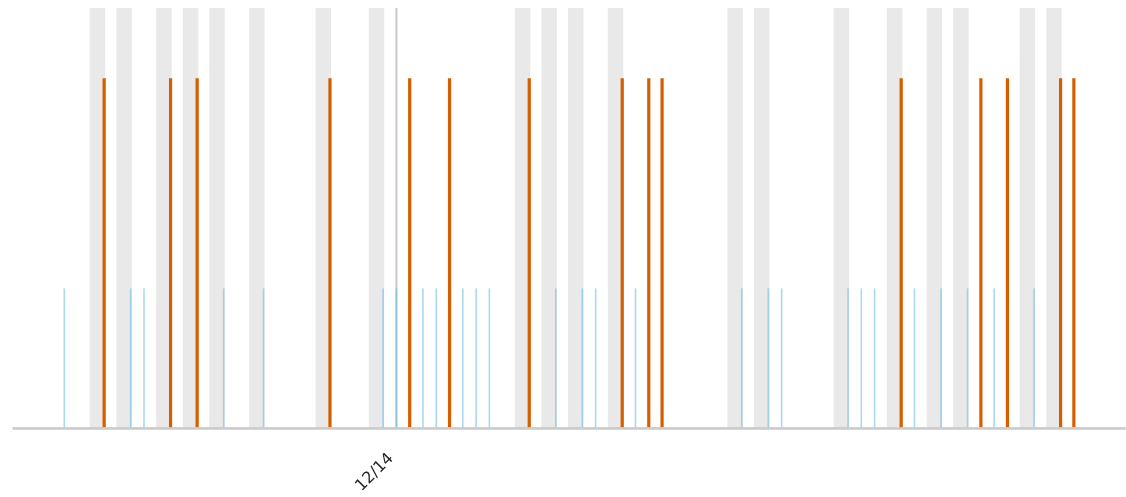


Per-patient Event Timelines — headache (patients 31-36)

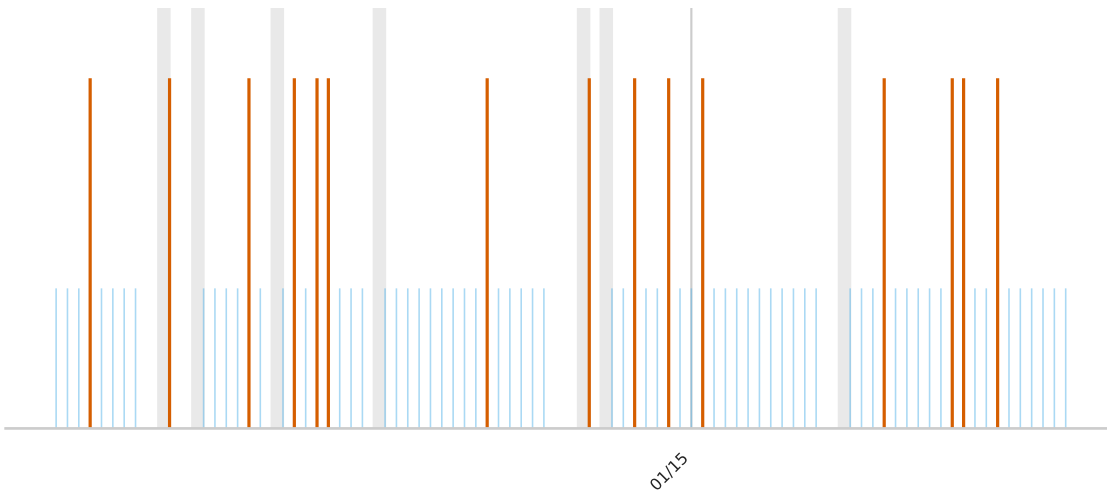
CMC-0002 | 15 events | 21.4%



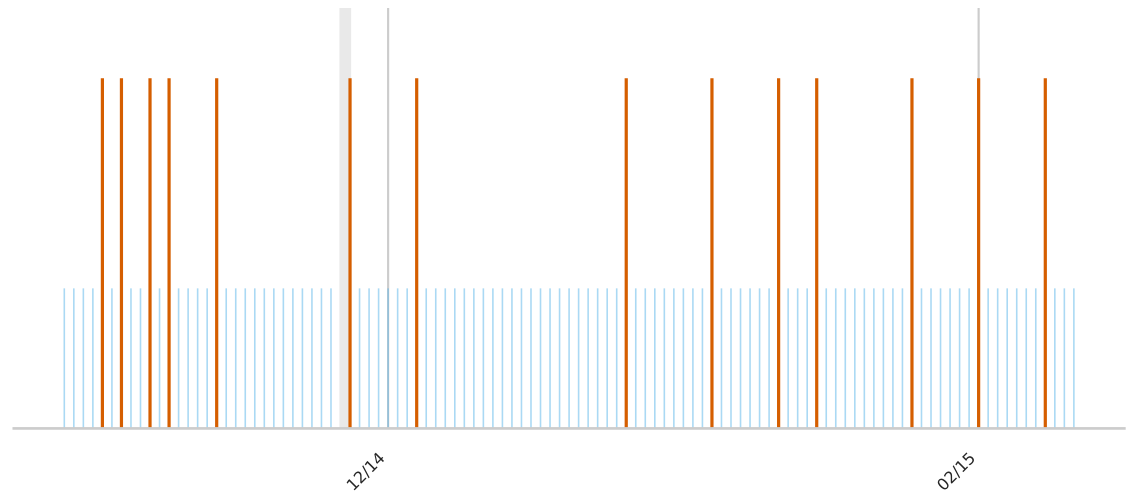
DHA-0044 | 15 events | 35.7%



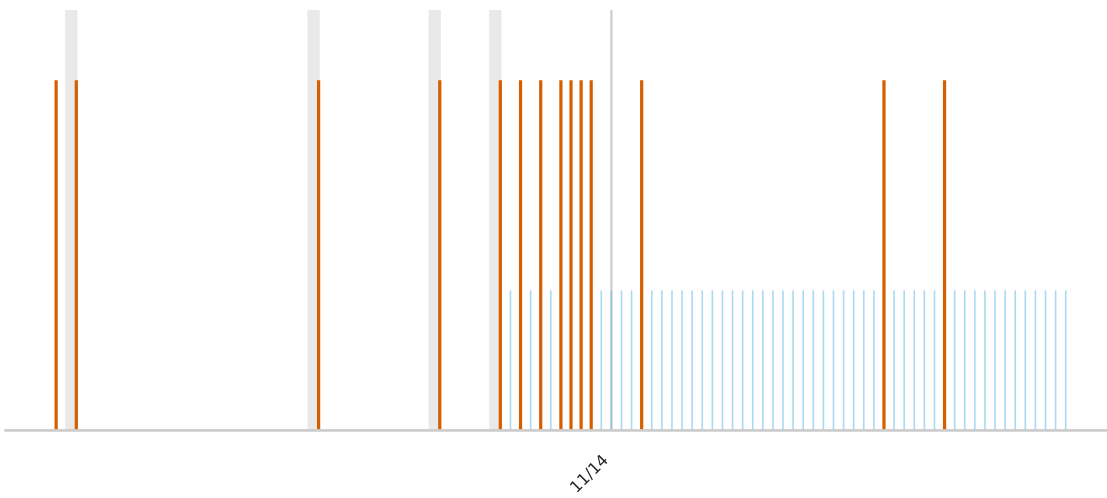
DHA-0050 | 15 events | 19.2%



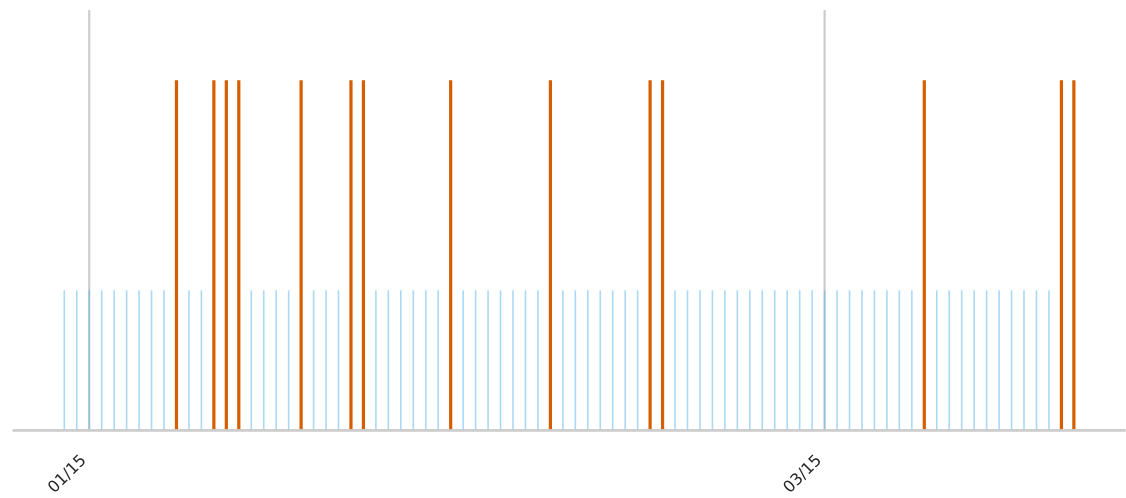
DHA-0041 | 14 events | 13.2%



CMC-0011 | 14 events | 23.0%

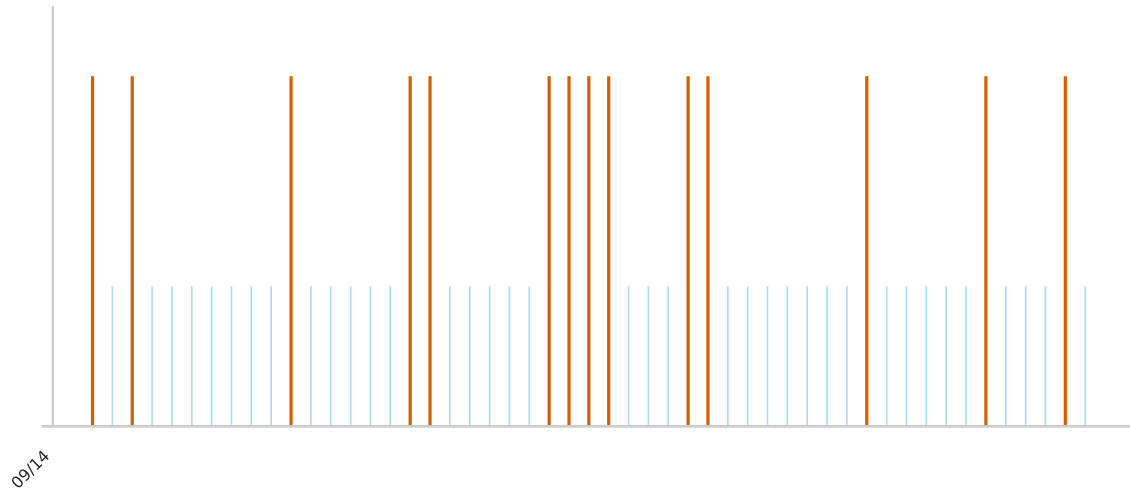


DHA-0061 | 14 events | 17.1%

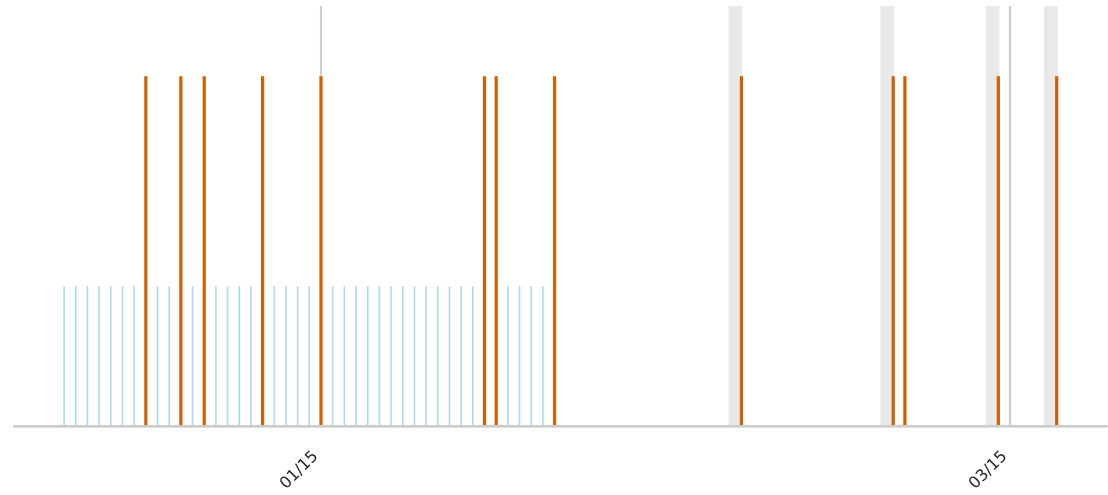


Per-patient Event Timelines — headache (patients 37-42)

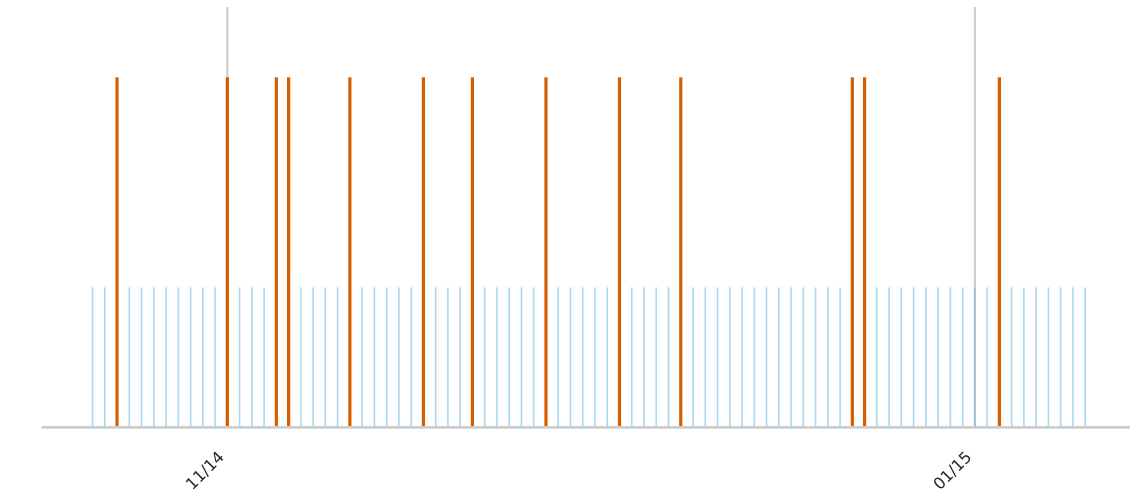
DHA-0015 | 14 events | 27.5%



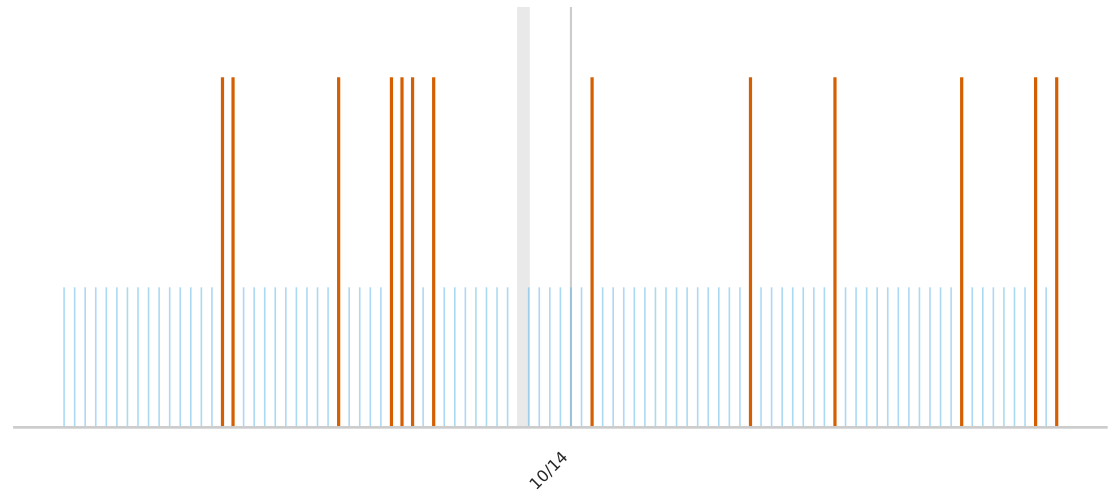
DHA-0055 | 13 events | 27.1%



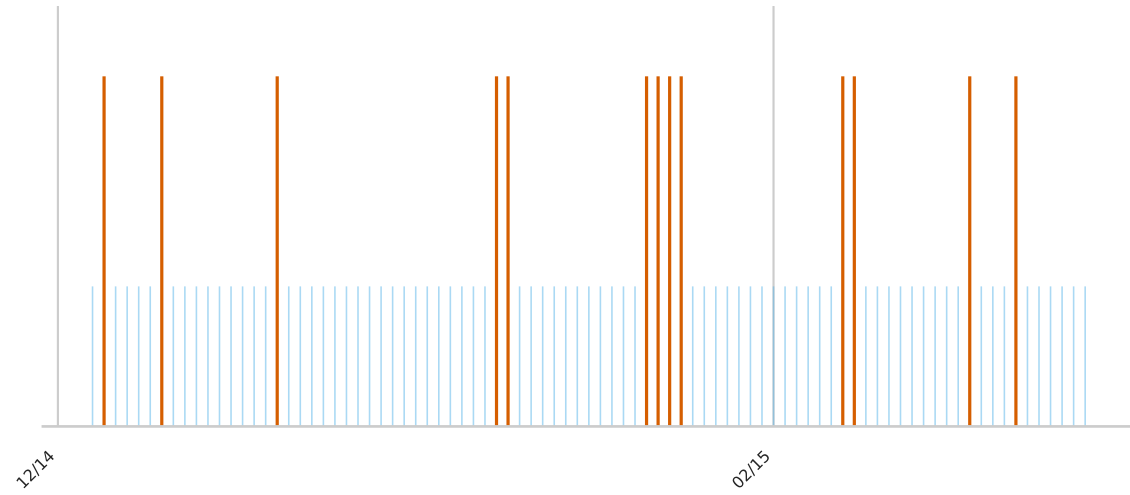
DHA-0039 | 13 events | 15.9%



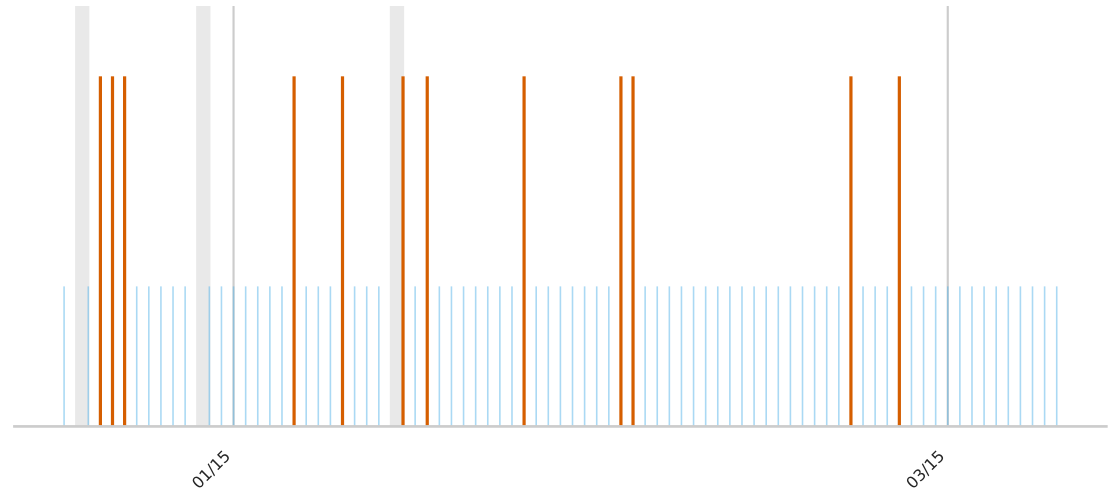
DHA-0002 | 13 events | 13.8%



DHA-0053 | 13 events | 14.9%

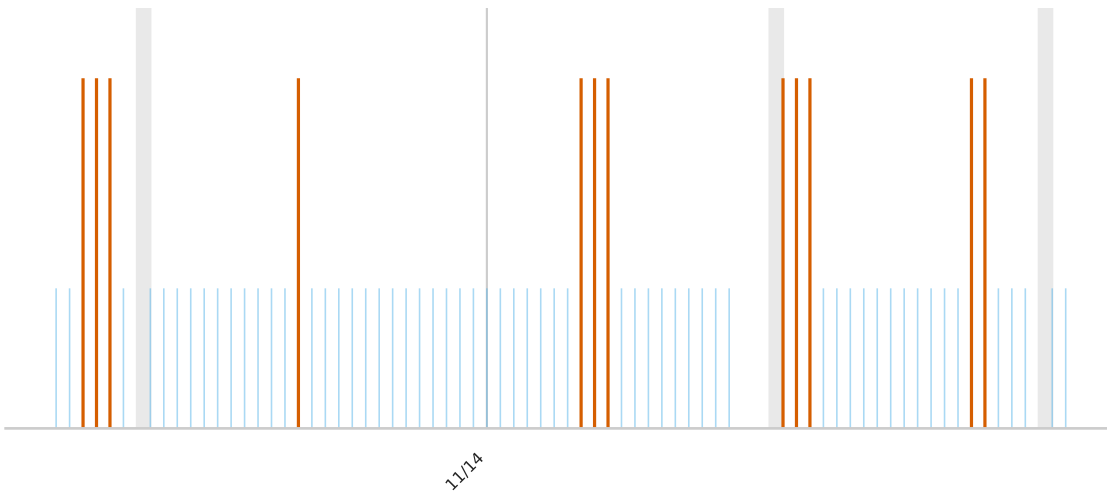


CMC-0068 | 12 events | 15.0%

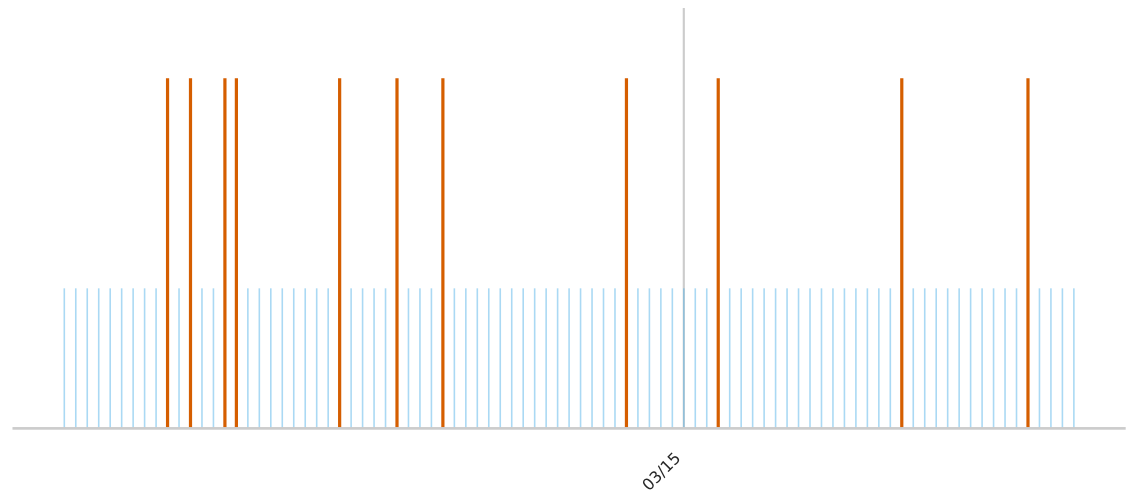


Per-patient Event Timelines — headache (patients 43-48)

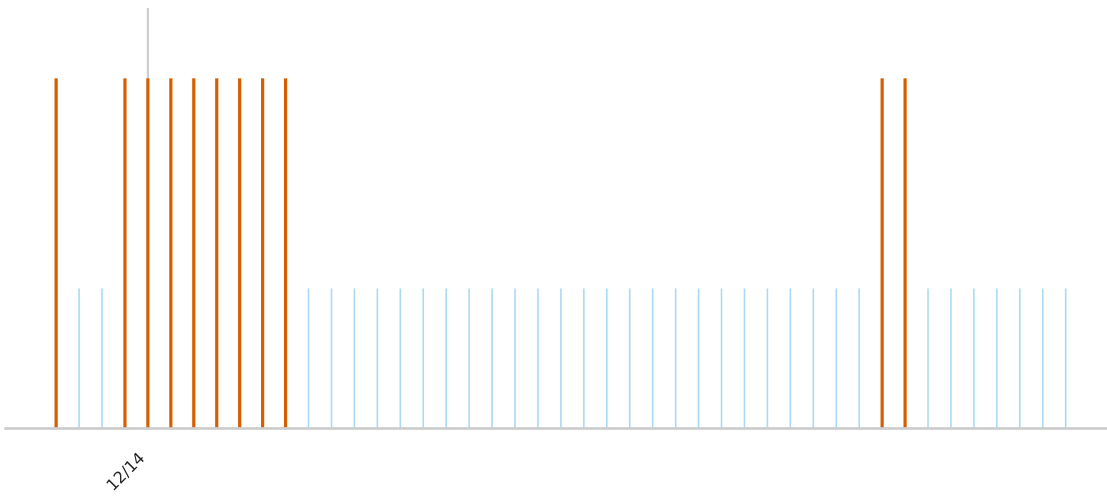
DHA-0023 | 12 events | 16.9%



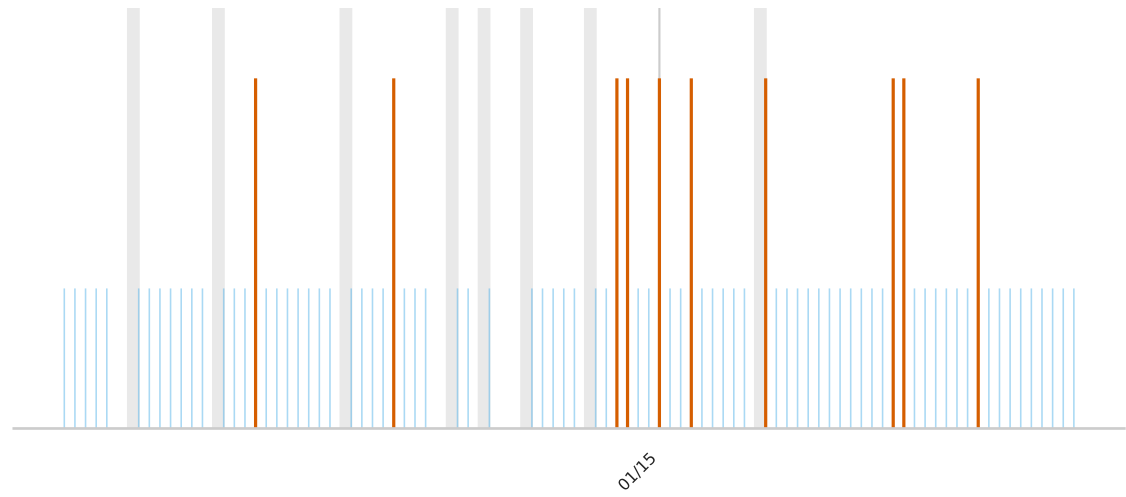
DHA-0063 | 11 events | 12.4%



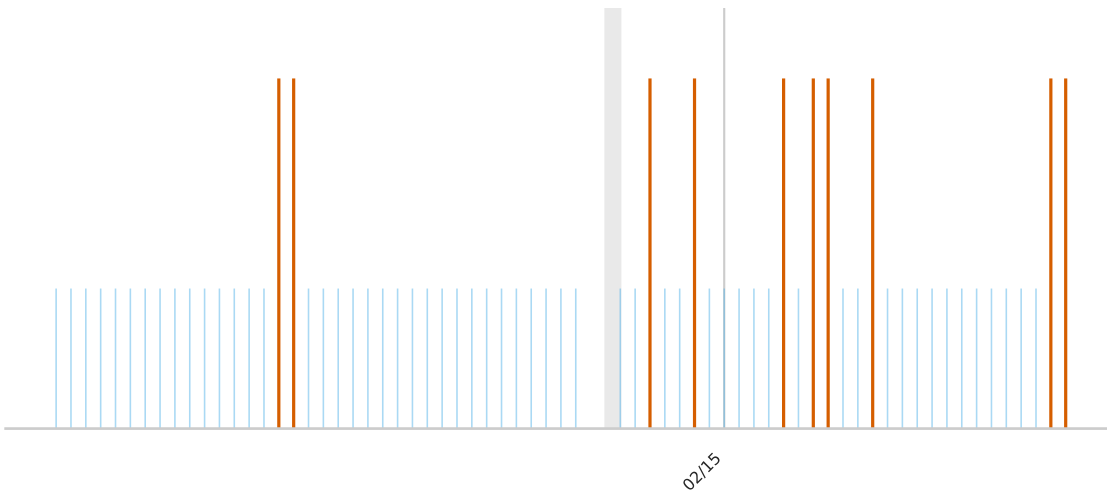
DHA-0051 | 11 events | 24.4%



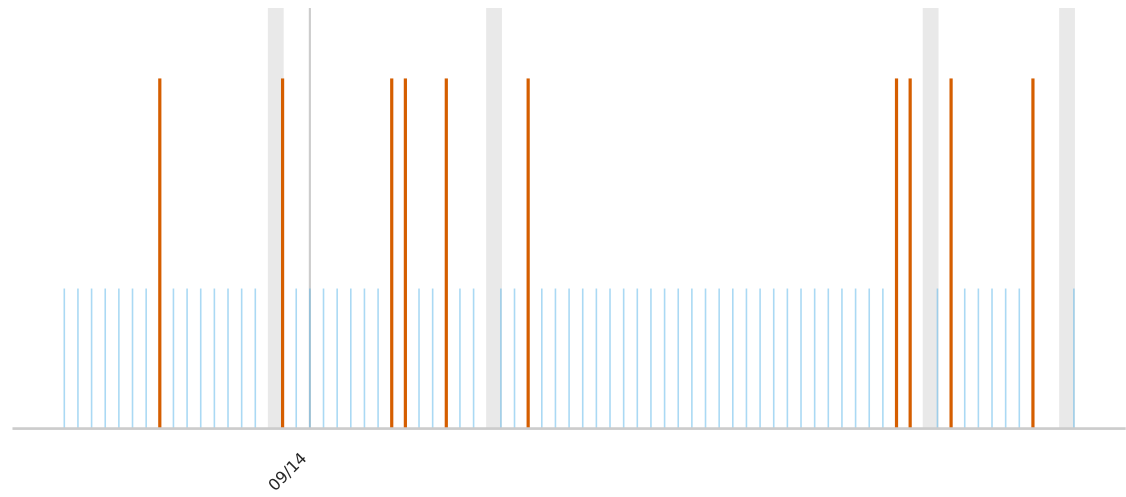
CMC-0048 | 10 events | 11.9%



CMC-0067 | 10 events | 14.9%

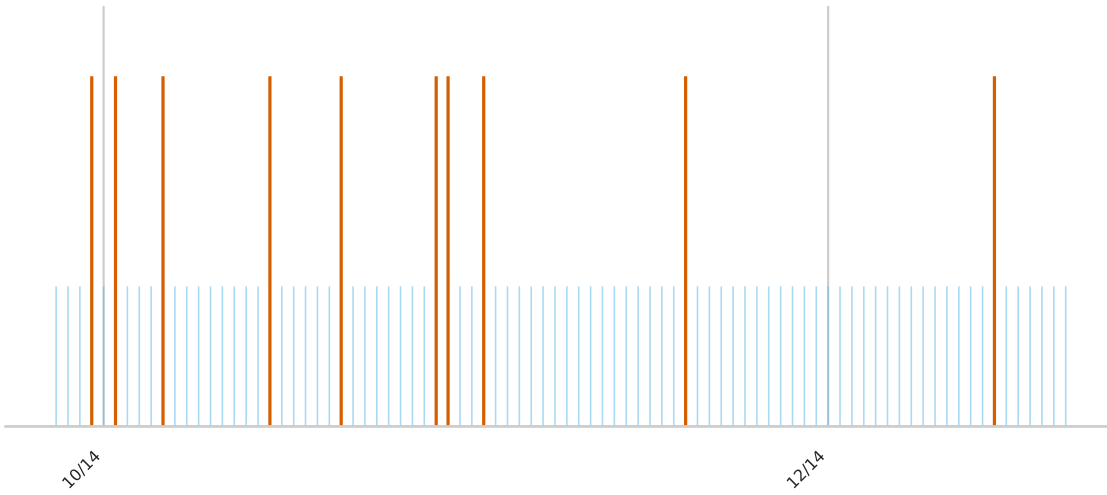


DHA-0004 | 10 events | 14.3%

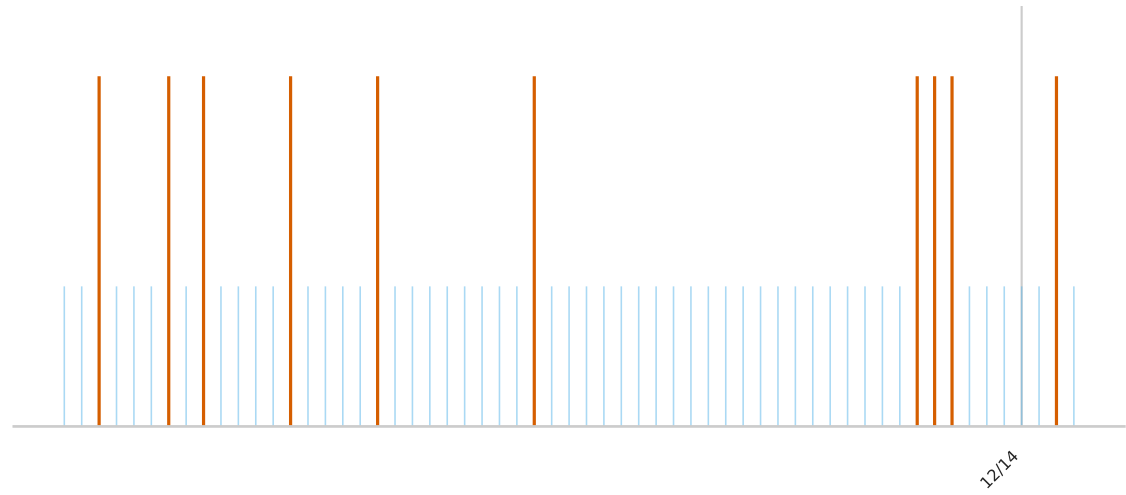


Per-patient Event Timelines — headache (patients 49-54)

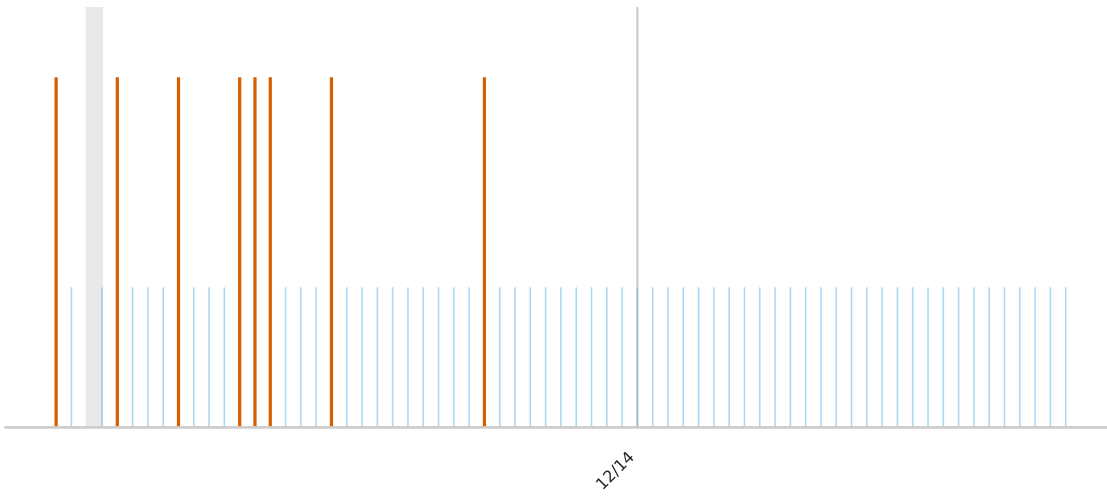
DHA-0022 | 10 events | 11.6%



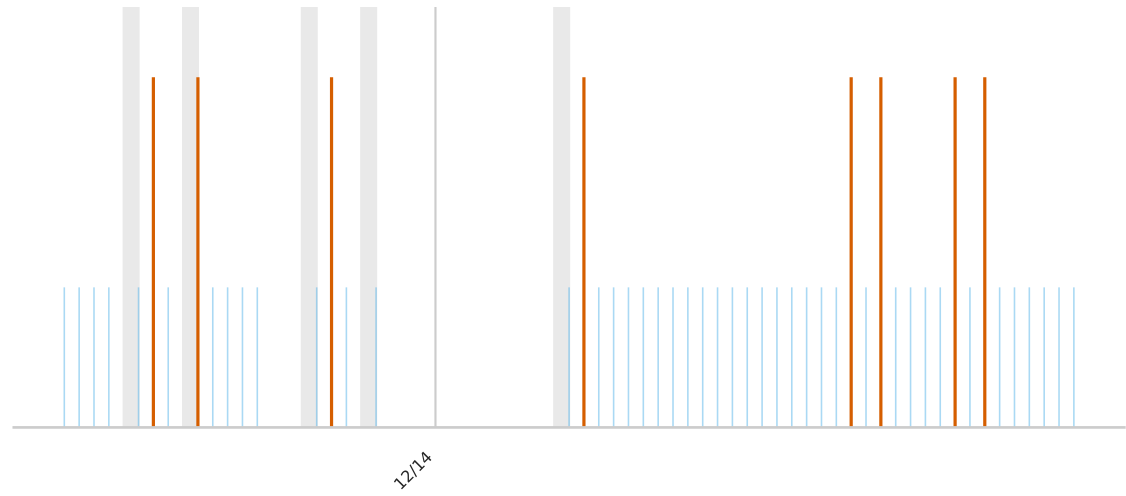
DHA-0029 | 10 events | 16.9%



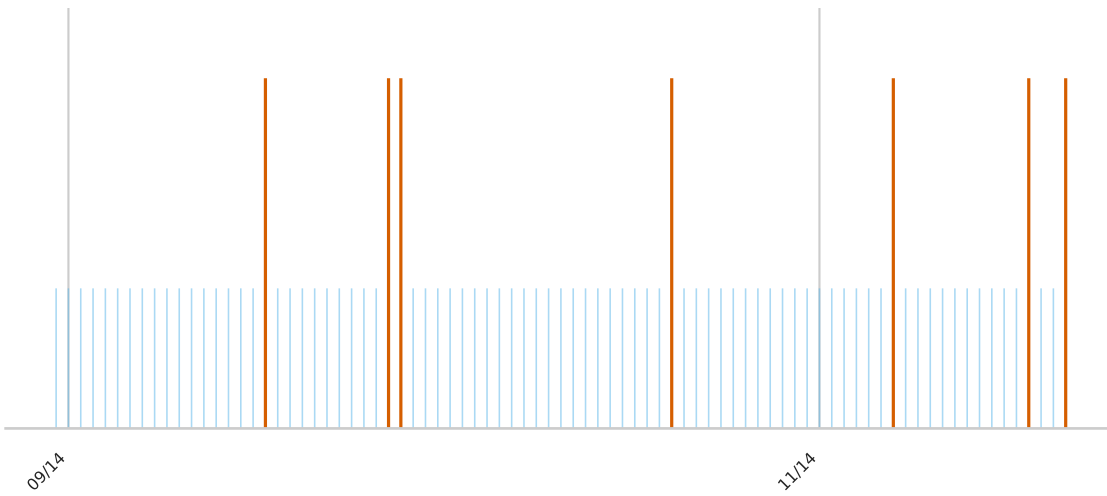
DHA-0040 | 8 events | 12.1%



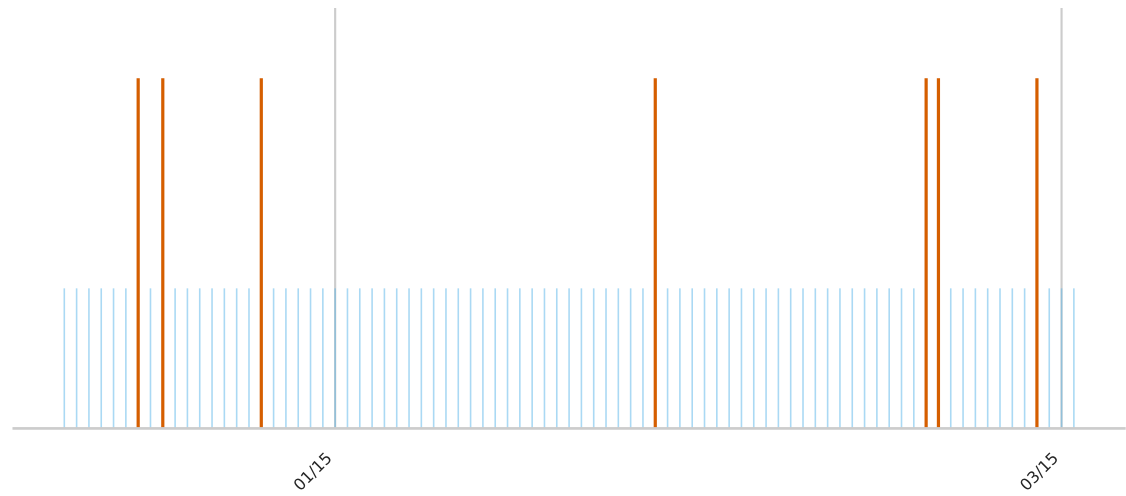
CMC-0049 | 8 events | 15.7%



DHA-0013 | 7 events | 8.4%

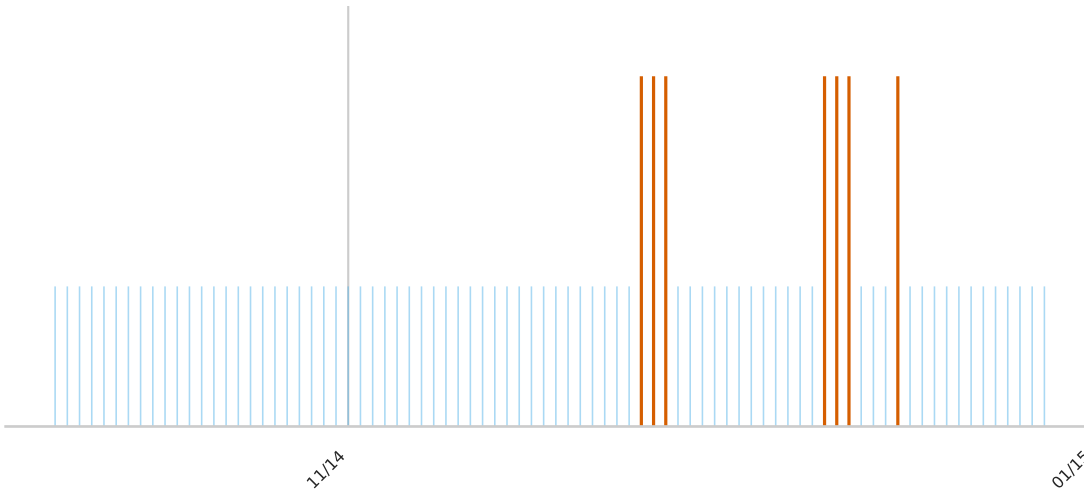


CMC-0064 | 7 events | 8.4%

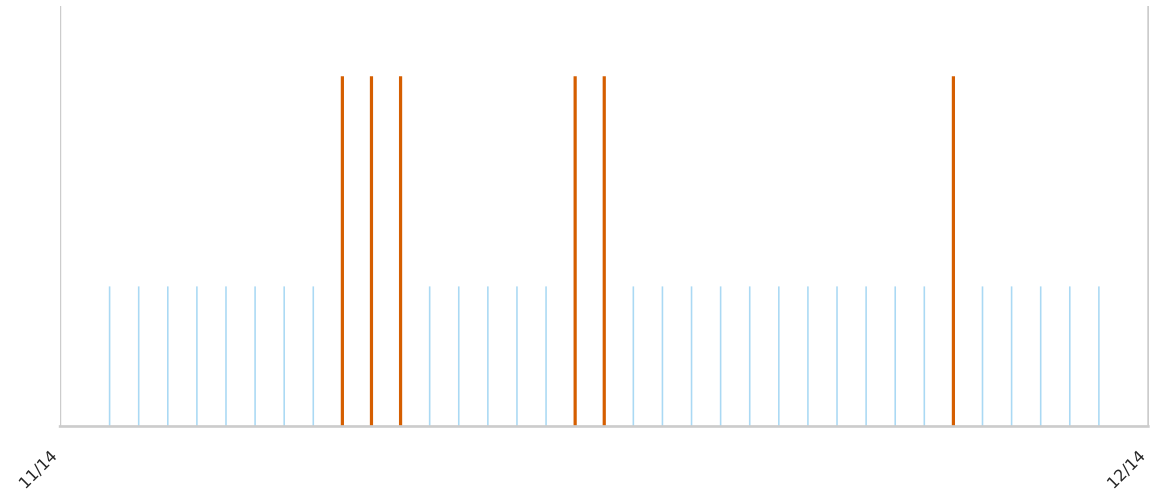


Per-patient Event Timelines — headache (patients 55-60)

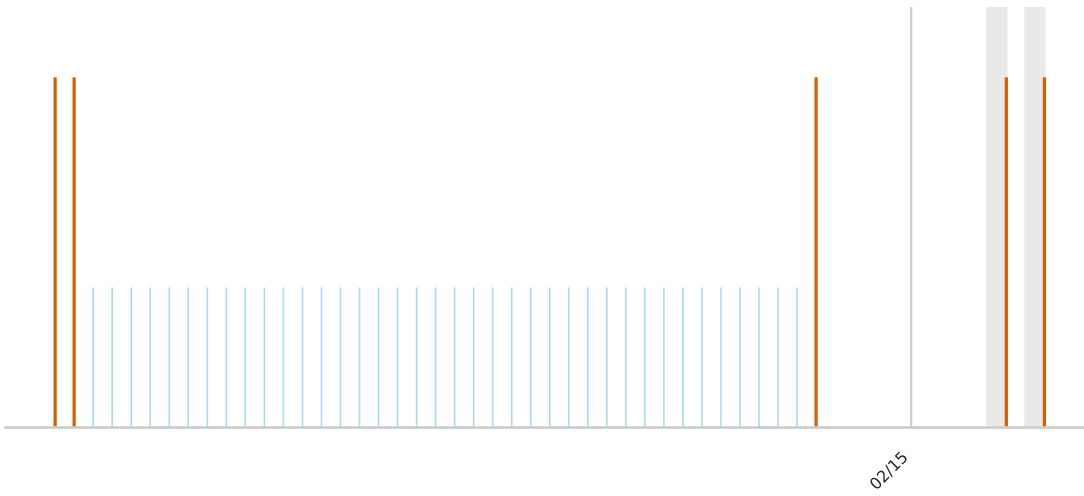
DHA-0032 | 7 events | 8.5%



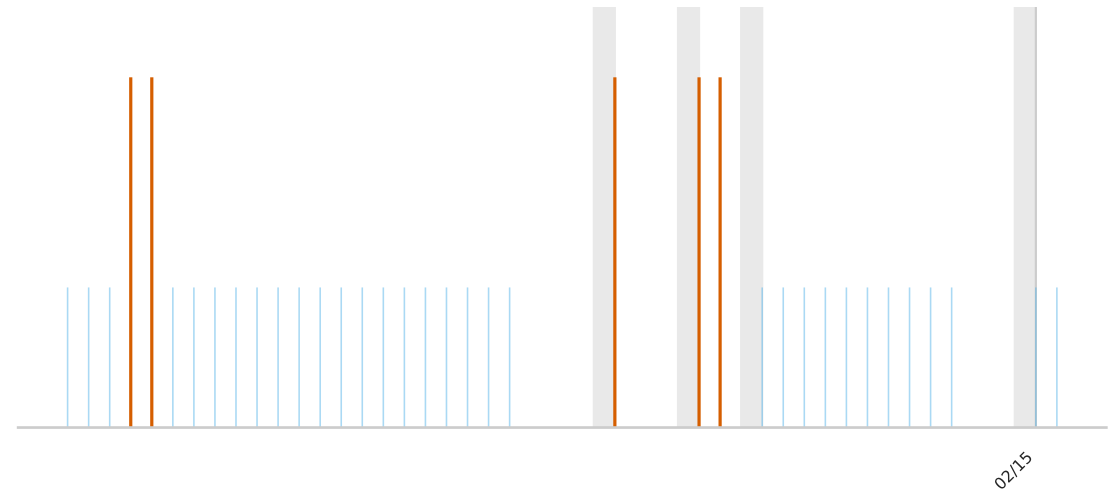
DHA-0031 | 6 events | 17.1%



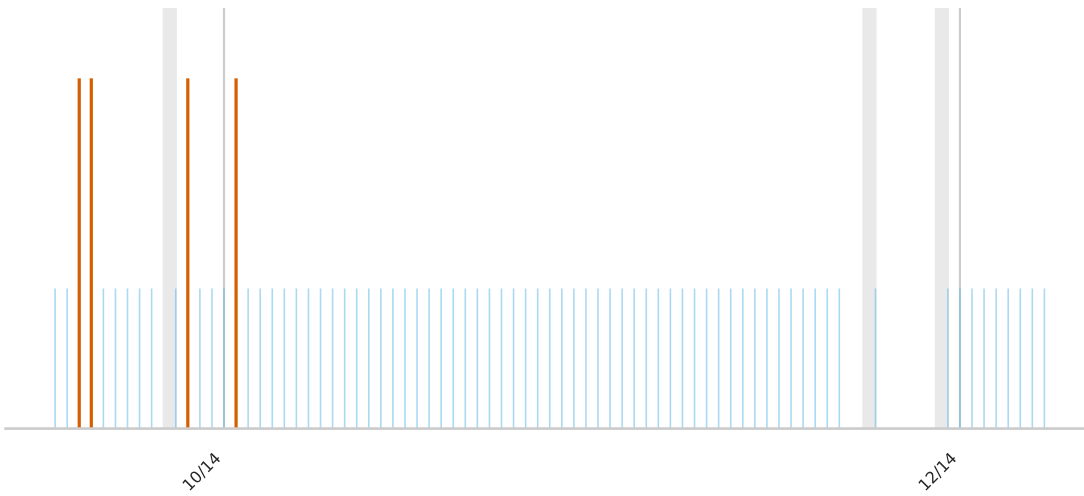
CMC-0069 | 5 events | 11.6%



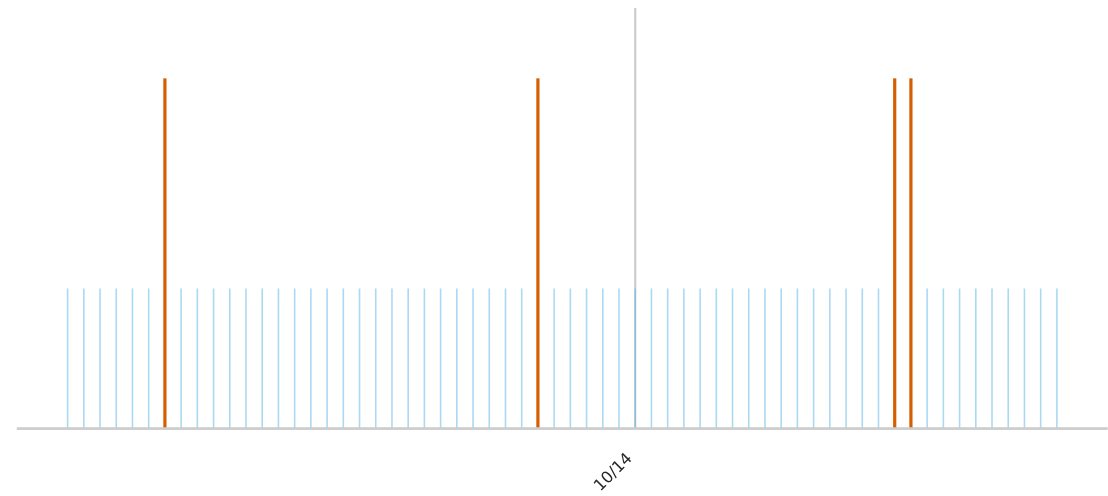
CMC-0066 | 5 events | 13.5%



CMC-0026 | 4 events | 5.3%



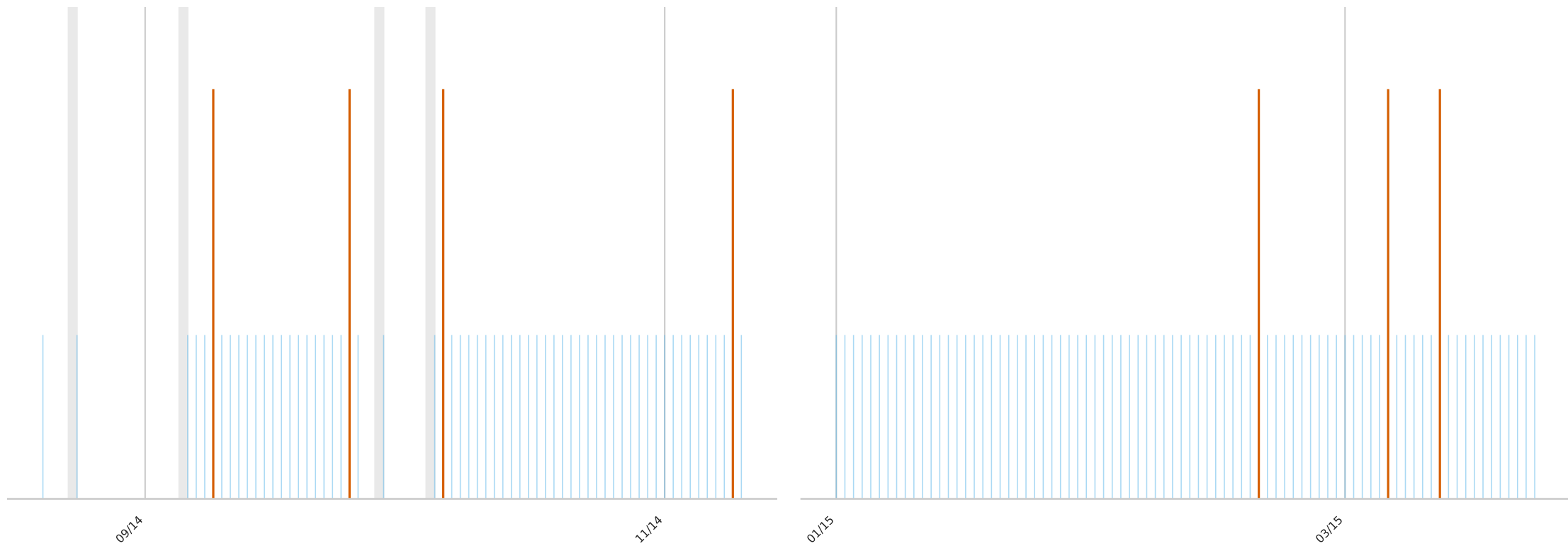
CMC-0004 | 4 events | 6.5%



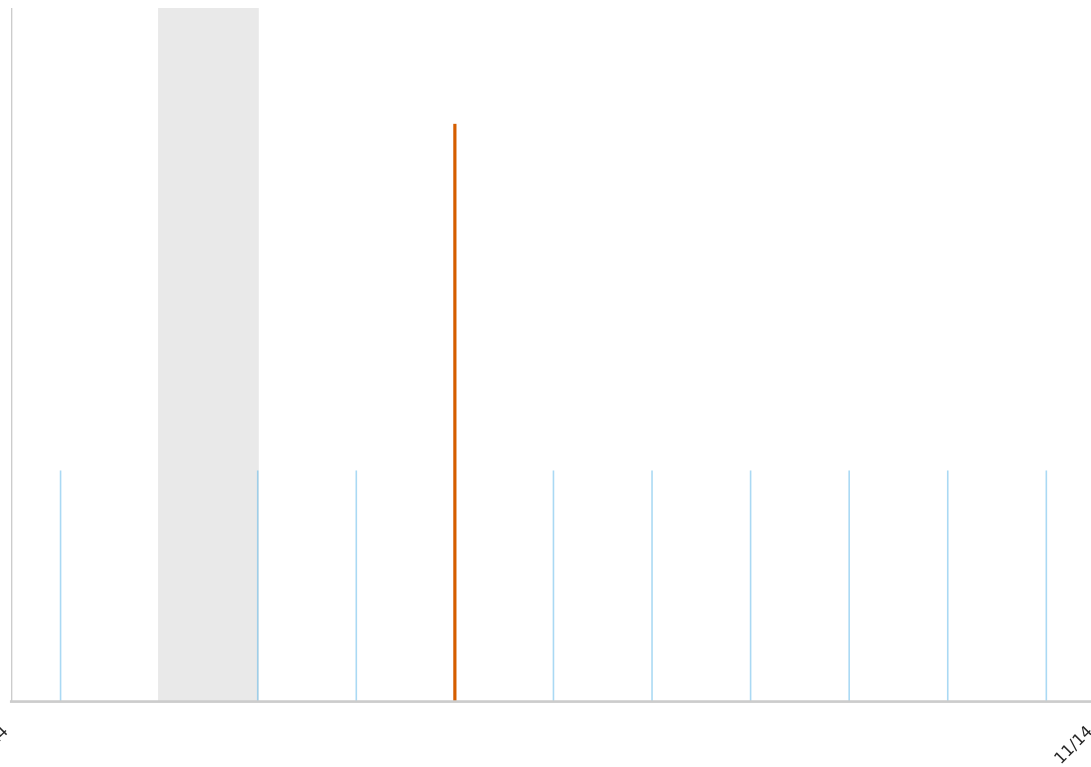
Per-patient Event Timelines — headache (patients 61-63)

CMC-0001 | 4 events | 6.6%

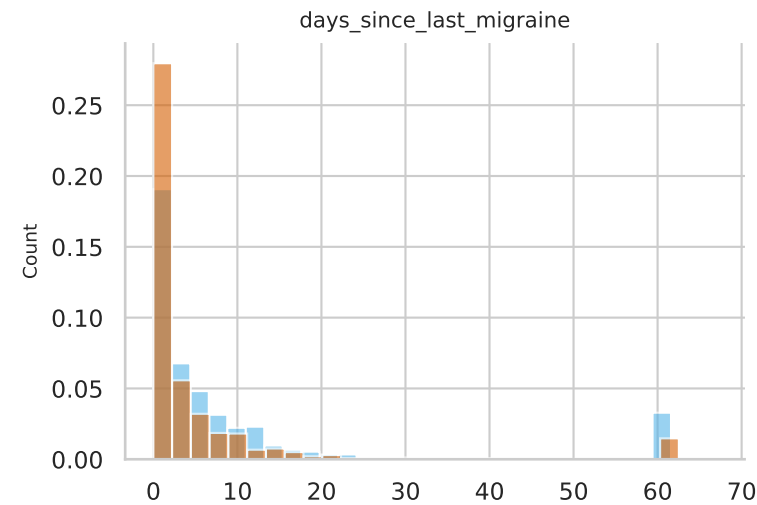
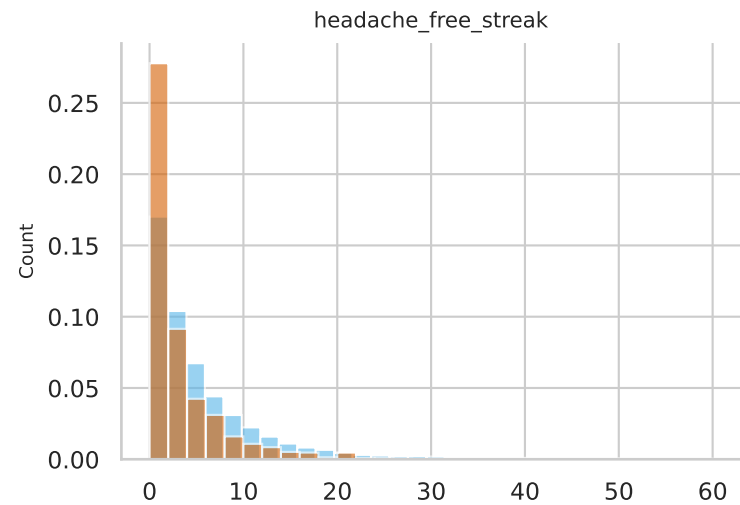
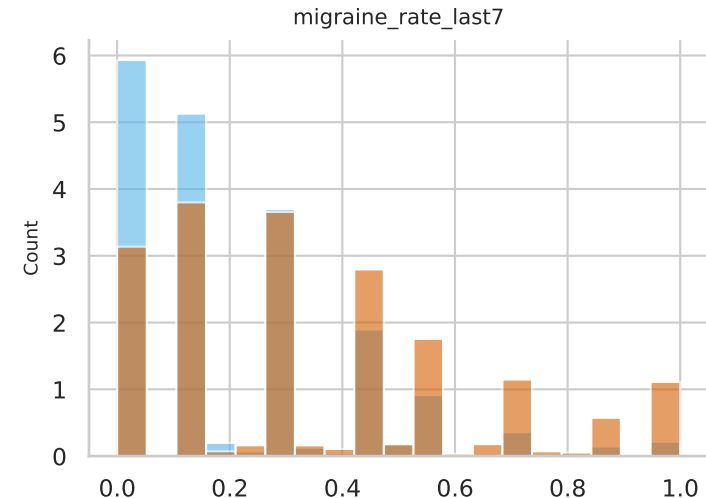
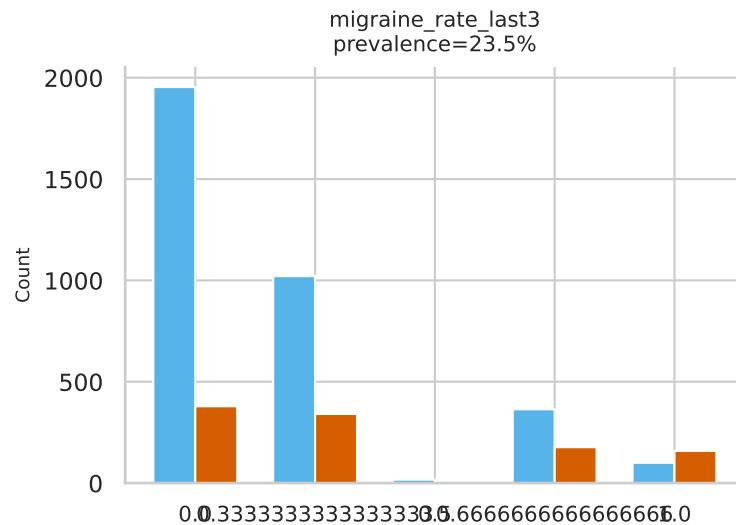
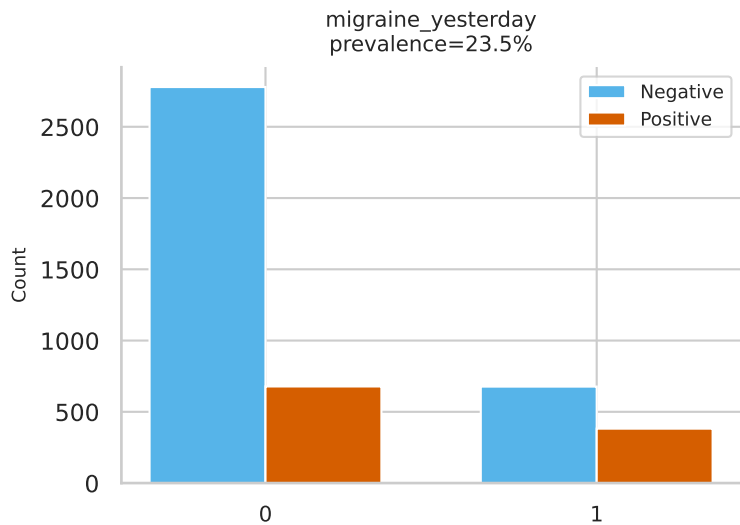
DHA-0062 | 3 events | 3.7%



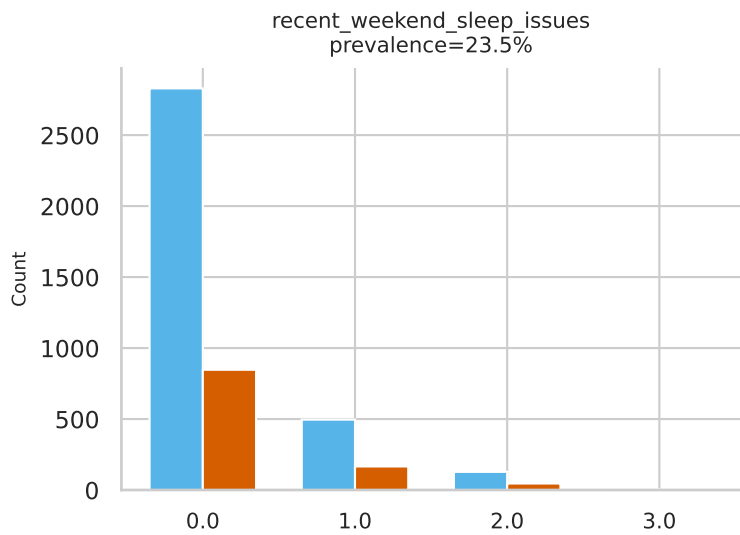
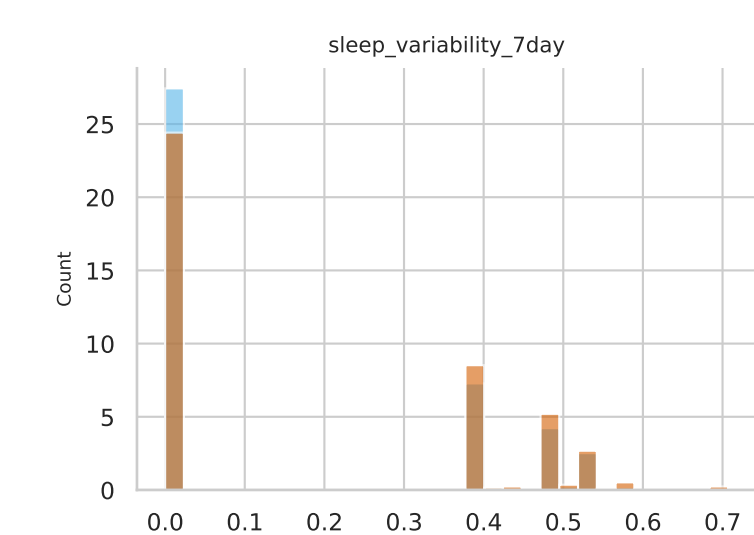
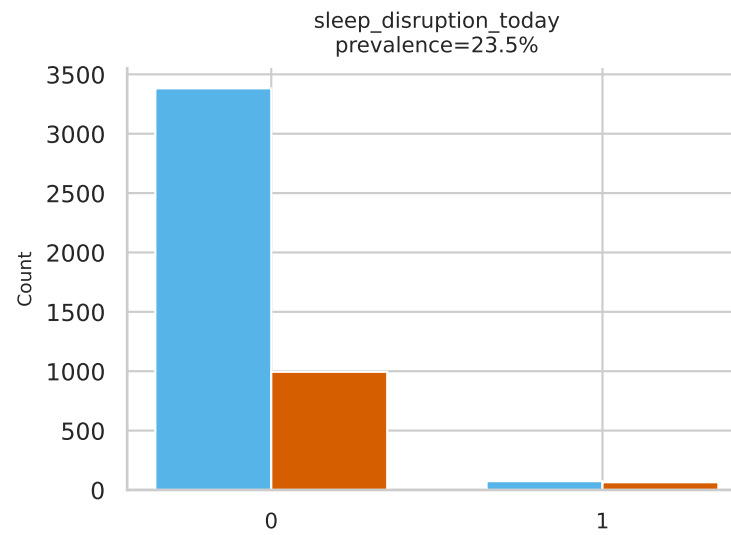
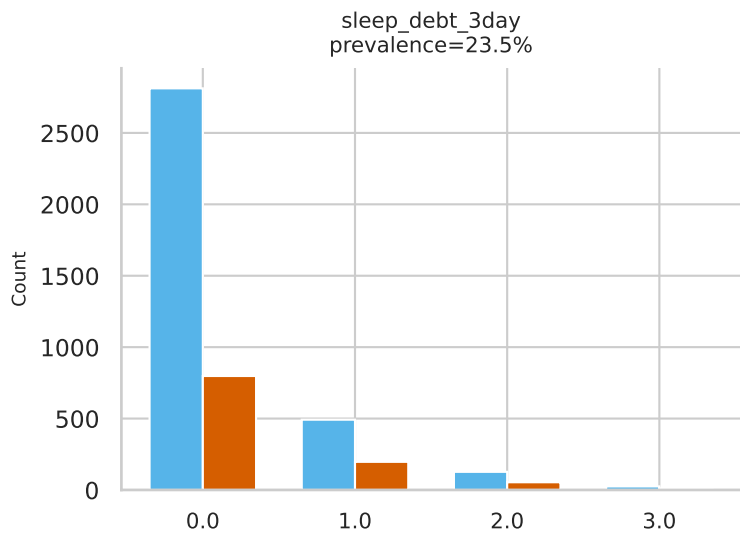
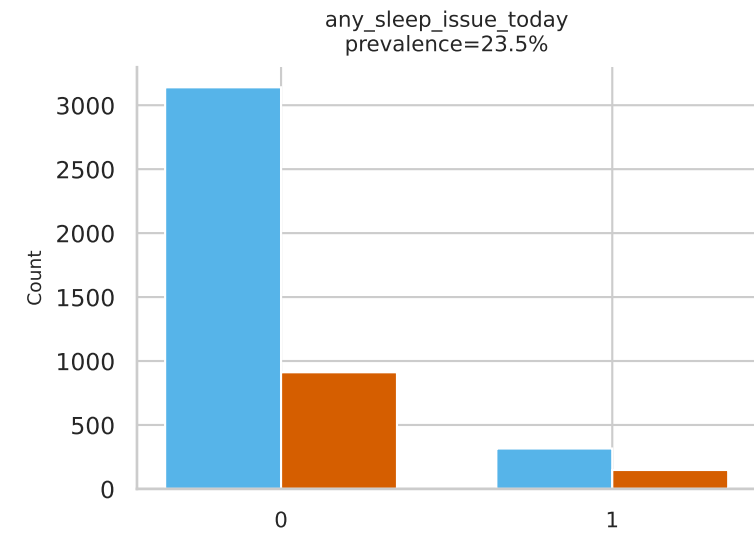
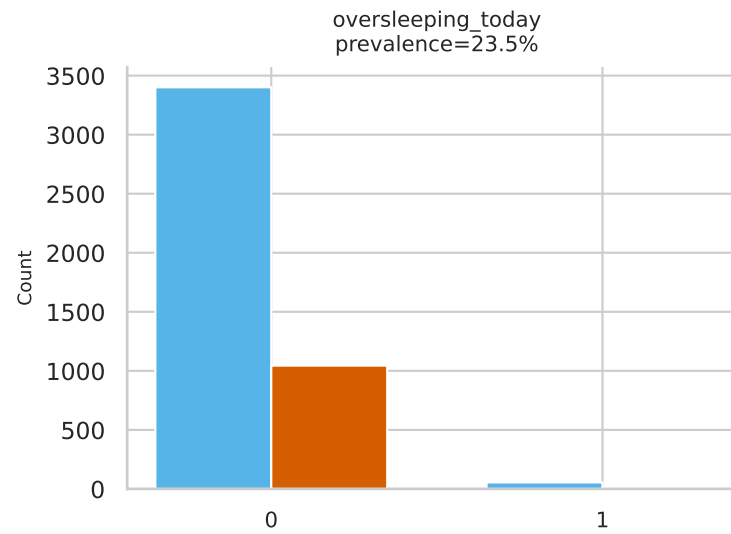
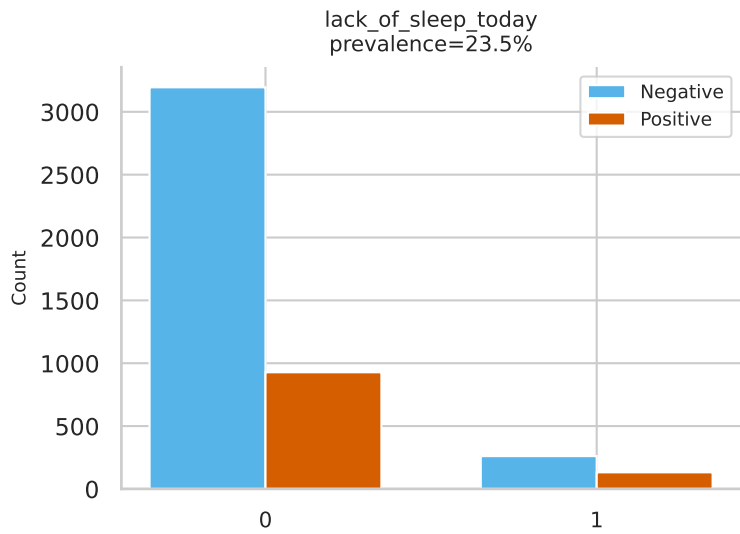
CMC-0050 | 1 events | 10.0%



Feature Distributions — Migraine History (headache)

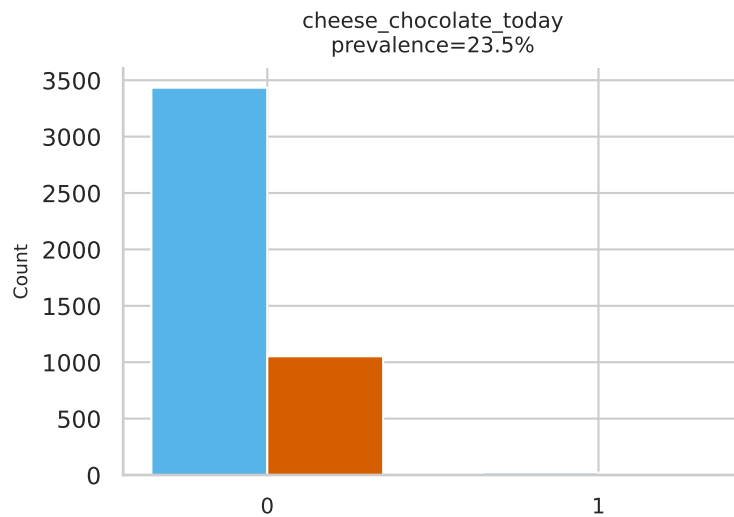
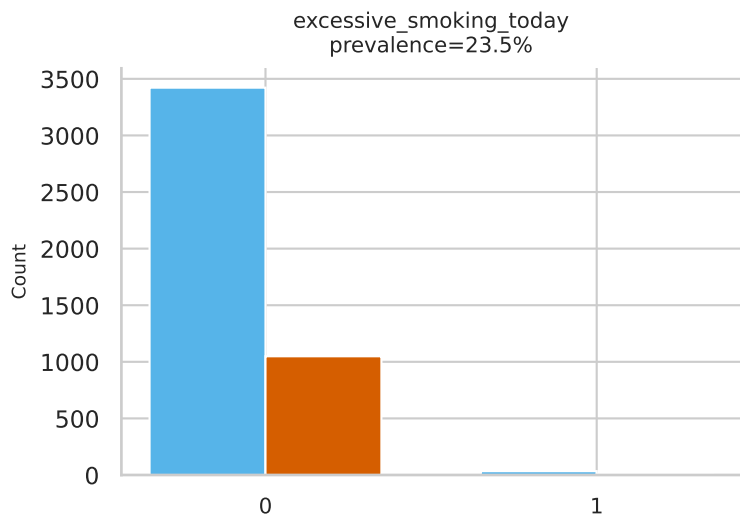
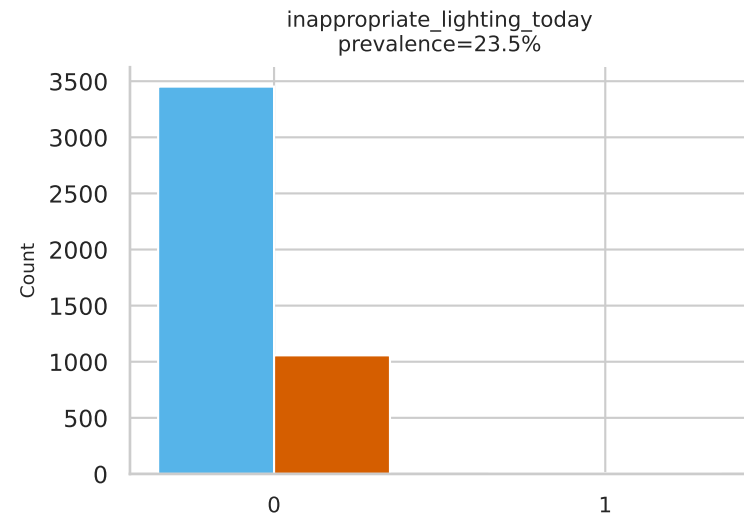
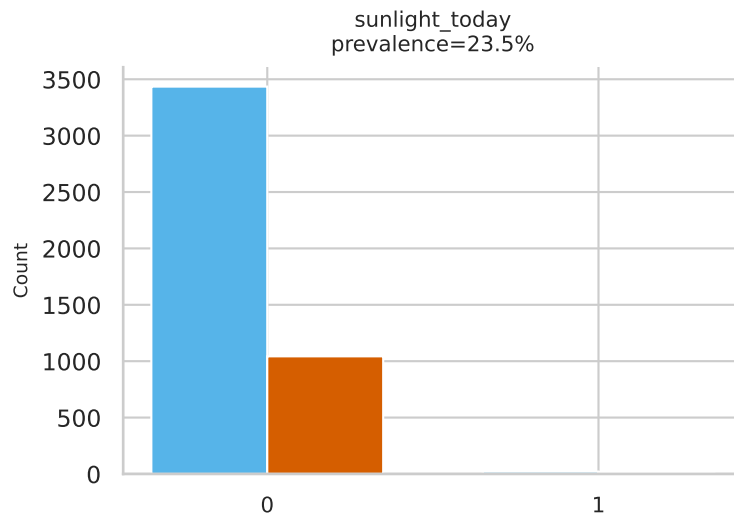
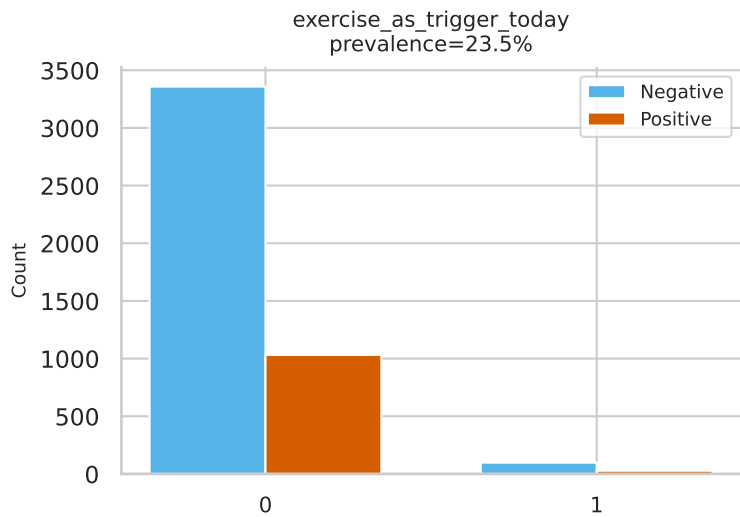


Feature Distributions — Sleep Features (headache)



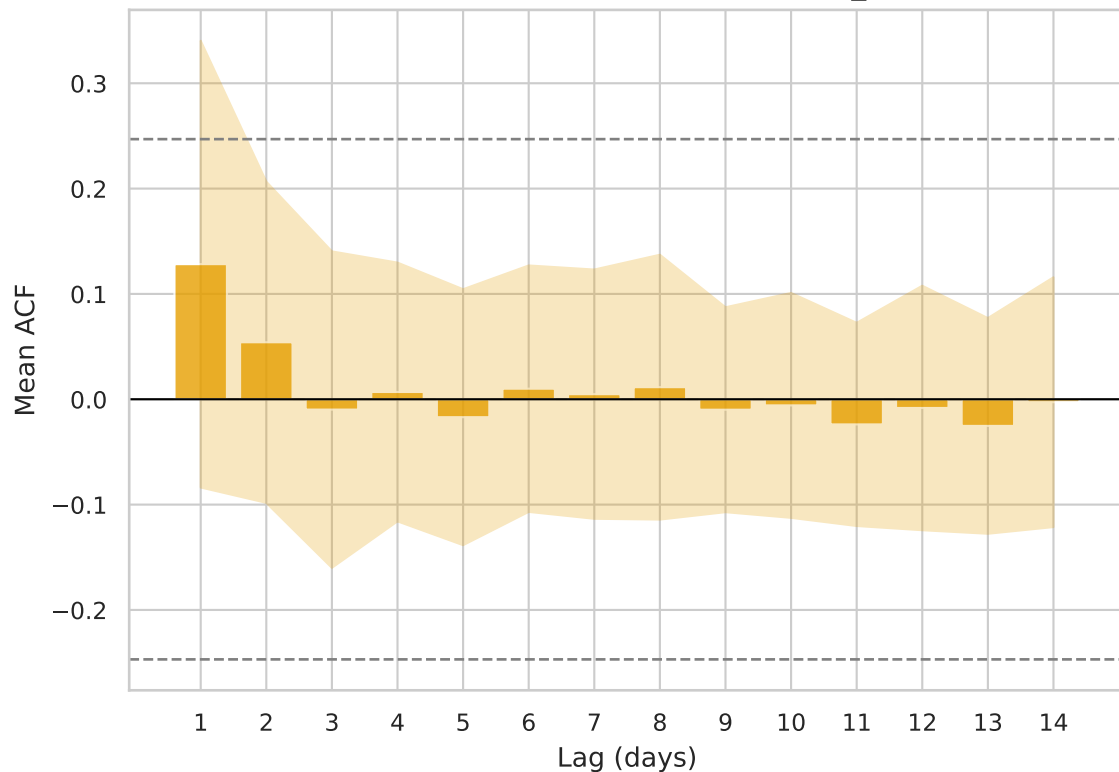


Feature Distributions — Low-Prevalence Restored Triggers (headache)

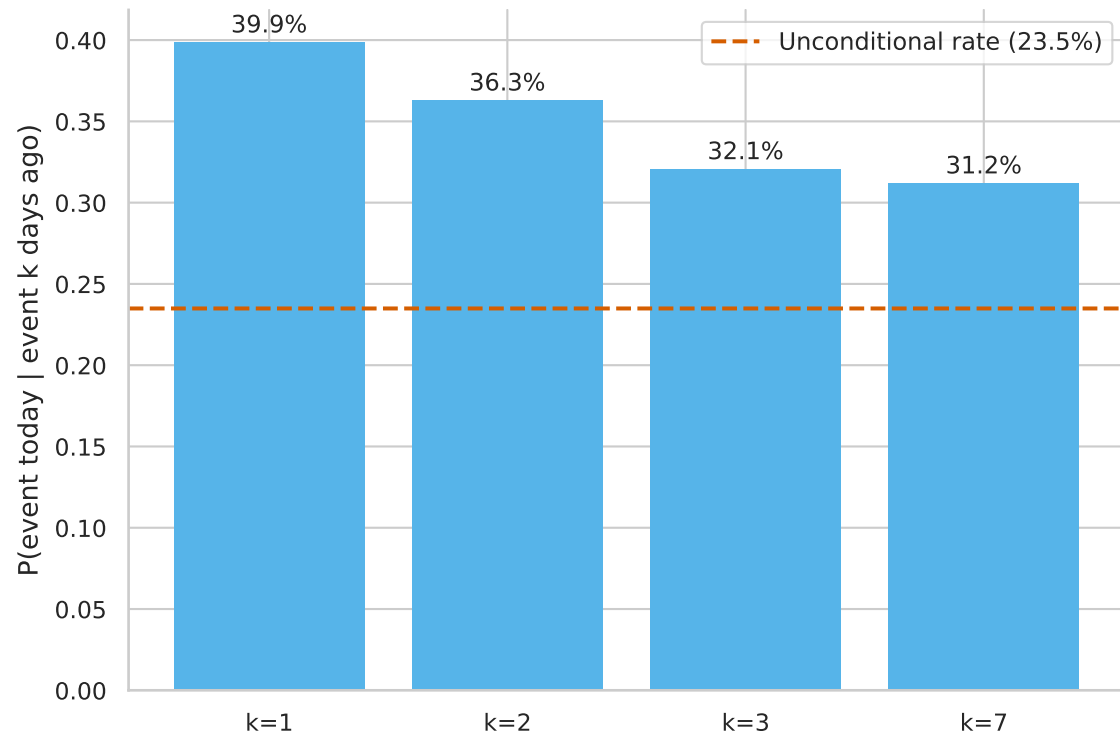


Temporal Dependence — headache

Population-averaged ACF of migraine_today



Conditional probability by lag



migraine_today shows weak lag-1 autocorrelation (ACF=0.13), suggesting dependence.